

Research in Computational Biology book

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Chapter 1

Prerequisites

This is a *sample* book written in **Markdown**. You can use anything that Pandoc's Markdown supports, e.g., a math equation $a^2 + b^2 = c^2$.

The **bookdown** package can be installed from CRAN or Github:

```
install.packages("bookdown")  
# or the development version  
# devtools::install_github("rstudio/bookdown")
```

Remember each Rmd file contains one and only one chapter, and a chapter is defined by the first-level heading #.

To compile this example to PDF, you need XeLaTeX. You are recommended to install TinyTeX (which includes XeLaTeX): <https://yihui.org/tinytex/>.

Chapter 2

Basics of R

2.1 Basic Steps

- R can be accessed by clicking an icon or entering the command “R” at the system command line, also referred to as Terminal
- This produces a console window or causes R to start up as an interactive program at the current terminal window
- R works fundamentally by a question-and-answer model: enter a line with a command and press Enter
- Then the program does something relevant
- In this course, we will use RStudio and a file format called R Markdown, which allows us to code using not only R, but bash and Python, two other languages we may refer to as the course progresses

2.2 Library ISwR

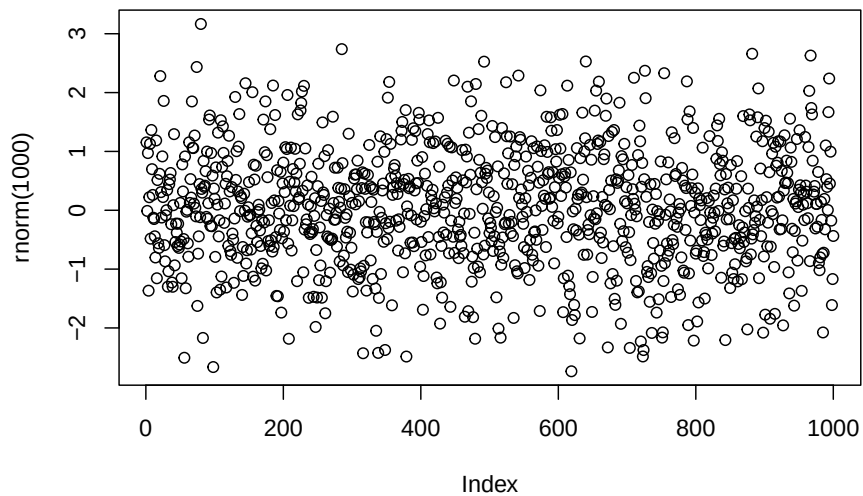
- This notebook used the book Introductory Statistics with R, by Peter Dalgaard as a reference
- For the book, R library ISwR (Introductory Statistics with R) can be freely downloaded
- All examples in the book used as reference should run provided that ISwR library is not only installed but also loaded into the current search path
- Library can be installed and loaded by typing the following command into an R chunk

```
install.packages("ISwR")  
library("ISwR")
```

2.3 Plot Random Numbers

- For a first impression of what R can do, let's try plotting a graph
- Need to insert the R chunk and use the plot function

```
plot(rnorm(1000))
```



2.4 A Potent Calculator

- R can process simple and complex arithmetic expression and produce a result for the user

```
2 + 2
```

```
## [1] 4
```

- R can also be used to do other standard calculations. Here is how to calculate e to the power of -2

```
exp(-2)
```

```
## [1] 0.1353353
```

- Other than the R chunks, these calculations can be made using the RStudio Console
- In RStudio, when using the R Markdown format (.Rmd), we can insert R, Bash (terminal) and Python chunks
- Proceed to illustrate that in the RStudio IDE

2.5 Assignments

- Assignments are made based on the necessity to store the results of calculations and use these results in downstream processing steps in an entire algorithm or pipeline
- Like other languages, R has symbolic **variables**: names that can be used to represent values
- To assign the value 2 to variable x, we can enter the following command

```
x <- 2
```

- The character <- is called the assignment operator
- There is no immediate visible result, but from now on, x has the value 2 and can be used in subsequent arithmetic operations
- We can “ask” R which value is stored in the x variable

```
x
```

```
## [1] 2
```

2.6 Operations after assignment of variable

- Below, our variable x, is used to perform other calculations

```
x
```

```
## [1] 2
```

- Addition

```
x + x
```

```
## [1] 4
```

- Multiplication

```
5*x
```

```
## [1] 10
```

- Potentiation

```
x^3
```

```
## [1] 8
```

2.7 Vectorized Arithmetic

- It is not useful to use one number at a time to run statistics
- One strength of R is that it can handle entire data vectors as single objects
- A data vector is an array of numbers and a vector variable can be constructed like this

```
weight <- c(60, 72, 57, 90, 95, 72)
```

```
## To look at the vector variable, just type its name again  
weight
```

```
## [1] 60 72 57 90 95 72
```

- You can do calculations with vectors just like ordinary numbers, so long as they have the same length

- One exception to this rule that we will see will be when we use the mean of weights of persons (represented by $\bar{x}$)
- In that case, the mean will be one single number, which will be subtracted from each sample value
- Suppose the weight vector indicates the weight of men in kilograms
- One simple formula to indicate whether a person is obese or not, is the body mass index (BMI)
- BMI is calculated by dividing the person's weight by the square of their height, in meters
- Therefore, in R, we need to have a vector with the height values to calculate the bmi vector, containing the body-mass index for the individuals indicated in the weight vector

```
height <- c(1.75, 1.80, 1.65, 1.90, 1.74, 1.91)
bmi <- weight/height^2
bmi
```

```
## [1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
```

2.8 Calculate the Mean of the variable weight

- The mean is calculated by the sum of the observations divided by the total number of observations

$$\bar{x} = \sum x_i / n$$

- Let's calculate the mean of the variable weight

```
sum(weight)
```

```
## [1] 446
```

```
sum(weight)/length(weight)
```

```
## [1] 74.33333
```

2.9 Calculate the Standard-Deviation of the variable weight

- Standard-deviation can be calculated with the following equation

$$SD = \sqrt{(\sum (x_i - \bar{x})^2) / (n - 1)}$$

- $\bar{x}$, the mean of variable weight, can be calculated using the sum and length of variable weight

```
xbar <- sum(weight)/length(weight)
xbar
```

```
## [1] 74.33333
```

- Now we can calculate the difference of each replicate in the weight variable and the mean of the weight variable, one by one

```
weight - xbar
```

```
## [1] -14.333333 -2.333333 -17.333333 15.666667 20.666667 -2.333333
```

- Notice how R uses $\bar{x}$, which has length one, to calculate the new weight - $\bar{x}$ data vector
- $\bar{x}$ is recycled and subtracted from every element in the weight variable (weight data vector)

2.10 Calculation of the Standard Deviation

- Calculate the squared deviations

```
(weight - xbar)^2
```

```
## [1] 205.444444 5.444444 300.444444 245.444444 427.111111 5.444444
```

- Calculate the sum of squared deviations

```
sum((weight - xbar)^2)
```

```
## [1] 1189.333
```

Calculate the standard deviation

```
sqrt(  
  sum(  
    (weight - xbar)^2/  
    (length(weight)-1)  
  )  
)
```

```
## [1] 15.42293
```

2.11 Standard Statistical Procedures

- It is a standard medical practice to assess whether a person is obese or not using validated scientific criteria
- As a simple procedure to show this concept, let's assume that an individual with a normal weight should have a BMI in the range 20-25
- We want to know if our data deviates from the normal range of BMI
- In R, this can be done using a statistical test called t-test
- You do not need to understand what a t-test is, just remember that it is used to evaluate the distribution of sample values compared to the normal distribution
- You can use a one-sample t-test to assess whether the six persons' BMI can be assumed to have mean 22.5 given that they come from a normal distribution
- You can do that using the function `t.test`

2.12 Standard Statistical Procedures: t-test

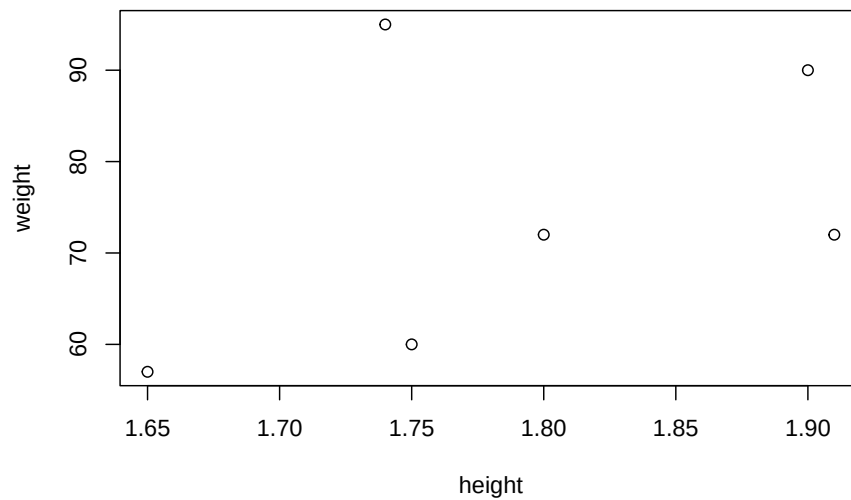
```
t.test(  
  bmi,  
  mu = 22.5  
)
```

```
##  
## One Sample t-test  
##  
## data:  bmi  
## t = 0.34488, df = 5, p-value = 0.7442  
## alternative hypothesis: true mean is not equal to 22.5  
## 95 percent confidence interval:  
##  18.41734 27.84791  
## sample estimates:  
## mean of x  
##  23.13262
```

2.13 Plot Graphics

- Let's now plot a scatterplot of the height and weight of individuals

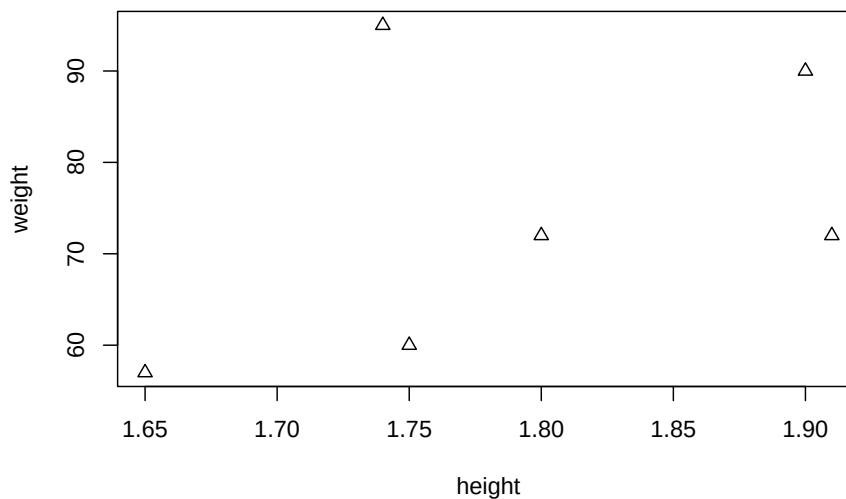
```
plot(height, weight)
```



2.14 Plot Graphics Modifying the plotting character

- You will frequently want to modify drawing of your graphs in various ways
- One way is using the parameter “plotting character”, `pch`

```
plot(height, weight, pch = 2)
```



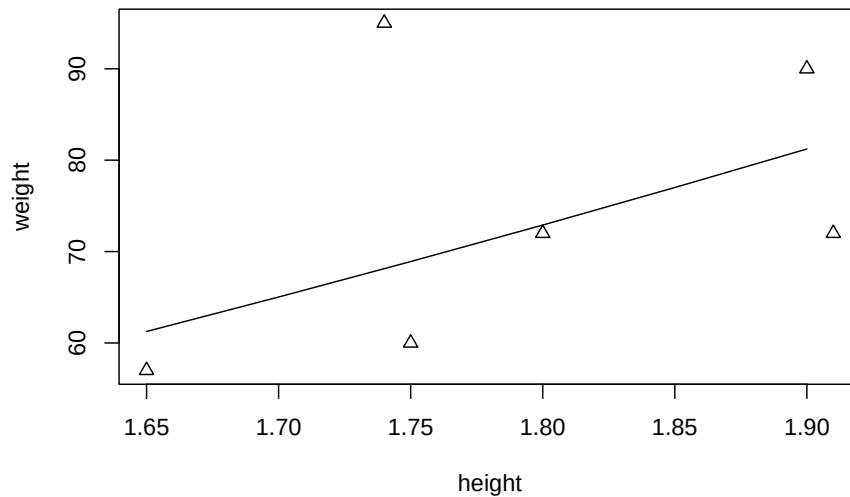
2.15 Plot an expected Line for weights at BMI of 22.5

- The term for weight at BMI of 22.5 is:

```
22.5*(height)^2
```

- This term can be used to compute the line of expected weights at a BMI of 22.5 using the `lines()` function
- As a note, we can't use the `lines()` function without first creating a plot in R.

```
plot(height, weight, pch = 2)
hh <- c(1.65, 1.70, 1.75, 1.80, 1.85, 1.90)
lines(hh, 22.5 * (hh)^2)
```



2.16 Vectors

- The weight and height vectors are called numeric vectors
- Besides numeric vectors, there are numeric and character vectors

2.17 Using other libraries

2.17.1 The iris data-frame

- Now we have a better understanding of what R can give us, let us use another library more commonly used datasets
- At times, it is possible that you will need to figure out different ways to install a library to use it

```
library(datasets)
```

- In the next chunk, we access the iris data, and look at a summary of the dataset

```
data(iris)
summary(iris)
```

```
##      Sepal.Length      Sepal.Width      Petal.Length      Petal.Width
##  Min.   :4.300      Min.   :2.000      Min.   :1.000      Min.   :0.100
##  1st Qu.:5.100      1st Qu.:2.800      1st Qu.:1.600      1st Qu.:0.300
##  Median :5.800      Median :3.000      Median :4.350      Median :1.300
##  Mean   :5.843      Mean   :3.057      Mean   :3.758      Mean   :1.199
##  3rd Qu.:6.400      3rd Qu.:3.300      3rd Qu.:5.100      3rd Qu.:1.800
##  Max.   :7.900      Max.   :4.400      Max.   :6.900      Max.   :2.500
##           Species
##  setosa    :50
##  versicolor:50
##  virginica :50
##
##
##
```

- Another form to look at the iris data-frame is typing its name

```
iris
```

```
##      Sepal.Length Sepal.Width Petal.Length Petal.Width      Species
## 1           5.1           3.5           1.4           0.2      setosa
## 2           4.9           3.0           1.4           0.2      setosa
## 3           4.7           3.2           1.3           0.2      setosa
## 4           4.6           3.1           1.5           0.2      setosa
## 5           5.0           3.6           1.4           0.2      setosa
## 6           5.4           3.9           1.7           0.4      setosa
## 7           4.6           3.4           1.4           0.3      setosa
## 8           5.0           3.4           1.5           0.2      setosa
## 9           4.4           2.9           1.4           0.2      setosa
## 10          4.9           3.1           1.5           0.1      setosa
## 11          5.4           3.7           1.5           0.2      setosa
## 12          4.8           3.4           1.6           0.2      setosa
## 13          4.8           3.0           1.4           0.1      setosa
## 14          4.3           3.0           1.1           0.1      setosa
## 15          5.8           4.0           1.2           0.2      setosa
```

## 16	5.7	4.4	1.5	0.4	setosa
## 17	5.4	3.9	1.3	0.4	setosa
## 18	5.1	3.5	1.4	0.3	setosa
## 19	5.7	3.8	1.7	0.3	setosa
## 20	5.1	3.8	1.5	0.3	setosa
## 21	5.4	3.4	1.7	0.2	setosa
## 22	5.1	3.7	1.5	0.4	setosa
## 23	4.6	3.6	1.0	0.2	setosa
## 24	5.1	3.3	1.7	0.5	setosa
## 25	4.8	3.4	1.9	0.2	setosa
## 26	5.0	3.0	1.6	0.2	setosa
## 27	5.0	3.4	1.6	0.4	setosa
## 28	5.2	3.5	1.5	0.2	setosa
## 29	5.2	3.4	1.4	0.2	setosa
## 30	4.7	3.2	1.6	0.2	setosa
## 31	4.8	3.1	1.6	0.2	setosa
## 32	5.4	3.4	1.5	0.4	setosa
## 33	5.2	4.1	1.5	0.1	setosa
## 34	5.5	4.2	1.4	0.2	setosa
## 35	4.9	3.1	1.5	0.2	setosa
## 36	5.0	3.2	1.2	0.2	setosa
## 37	5.5	3.5	1.3	0.2	setosa
## 38	4.9	3.6	1.4	0.1	setosa
## 39	4.4	3.0	1.3	0.2	setosa
## 40	5.1	3.4	1.5	0.2	setosa
## 41	5.0	3.5	1.3	0.3	setosa
## 42	4.5	2.3	1.3	0.3	setosa
## 43	4.4	3.2	1.3	0.2	setosa
## 44	5.0	3.5	1.6	0.6	setosa
## 45	5.1	3.8	1.9	0.4	setosa
## 46	4.8	3.0	1.4	0.3	setosa
## 47	5.1	3.8	1.6	0.2	setosa
## 48	4.6	3.2	1.4	0.2	setosa
## 49	5.3	3.7	1.5	0.2	setosa
## 50	5.0	3.3	1.4	0.2	setosa
## 51	7.0	3.2	4.7	1.4	versicolor
## 52	6.4	3.2	4.5	1.5	versicolor
## 53	6.9	3.1	4.9	1.5	versicolor
## 54	5.5	2.3	4.0	1.3	versicolor
## 55	6.5	2.8	4.6	1.5	versicolor
## 56	5.7	2.8	4.5	1.3	versicolor
## 57	6.3	3.3	4.7	1.6	versicolor
## 58	4.9	2.4	3.3	1.0	versicolor
## 59	6.6	2.9	4.6	1.3	versicolor
## 60	5.2	2.7	3.9	1.4	versicolor
## 61	5.0	2.0	3.5	1.0	versicolor

## 62	5.9	3.0	4.2	1.5 versicolor
## 63	6.0	2.2	4.0	1.0 versicolor
## 64	6.1	2.9	4.7	1.4 versicolor
## 65	5.6	2.9	3.6	1.3 versicolor
## 66	6.7	3.1	4.4	1.4 versicolor
## 67	5.6	3.0	4.5	1.5 versicolor
## 68	5.8	2.7	4.1	1.0 versicolor
## 69	6.2	2.2	4.5	1.5 versicolor
## 70	5.6	2.5	3.9	1.1 versicolor
## 71	5.9	3.2	4.8	1.8 versicolor
## 72	6.1	2.8	4.0	1.3 versicolor
## 73	6.3	2.5	4.9	1.5 versicolor
## 74	6.1	2.8	4.7	1.2 versicolor
## 75	6.4	2.9	4.3	1.3 versicolor
## 76	6.6	3.0	4.4	1.4 versicolor
## 77	6.8	2.8	4.8	1.4 versicolor
## 78	6.7	3.0	5.0	1.7 versicolor
## 79	6.0	2.9	4.5	1.5 versicolor
## 80	5.7	2.6	3.5	1.0 versicolor
## 81	5.5	2.4	3.8	1.1 versicolor
## 82	5.5	2.4	3.7	1.0 versicolor
## 83	5.8	2.7	3.9	1.2 versicolor
## 84	6.0	2.7	5.1	1.6 versicolor
## 85	5.4	3.0	4.5	1.5 versicolor
## 86	6.0	3.4	4.5	1.6 versicolor
## 87	6.7	3.1	4.7	1.5 versicolor
## 88	6.3	2.3	4.4	1.3 versicolor
## 89	5.6	3.0	4.1	1.3 versicolor
## 90	5.5	2.5	4.0	1.3 versicolor
## 91	5.5	2.6	4.4	1.2 versicolor
## 92	6.1	3.0	4.6	1.4 versicolor
## 93	5.8	2.6	4.0	1.2 versicolor
## 94	5.0	2.3	3.3	1.0 versicolor
## 95	5.6	2.7	4.2	1.3 versicolor
## 96	5.7	3.0	4.2	1.2 versicolor
## 97	5.7	2.9	4.2	1.3 versicolor
## 98	6.2	2.9	4.3	1.3 versicolor
## 99	5.1	2.5	3.0	1.1 versicolor
## 100	5.7	2.8	4.1	1.3 versicolor
## 101	6.3	3.3	6.0	2.5 virginica
## 102	5.8	2.7	5.1	1.9 virginica
## 103	7.1	3.0	5.9	2.1 virginica
## 104	6.3	2.9	5.6	1.8 virginica
## 105	6.5	3.0	5.8	2.2 virginica
## 106	7.6	3.0	6.6	2.1 virginica
## 107	4.9	2.5	4.5	1.7 virginica

## 108	7.3	2.9	6.3	1.8	virginica
## 109	6.7	2.5	5.8	1.8	virginica
## 110	7.2	3.6	6.1	2.5	virginica
## 111	6.5	3.2	5.1	2.0	virginica
## 112	6.4	2.7	5.3	1.9	virginica
## 113	6.8	3.0	5.5	2.1	virginica
## 114	5.7	2.5	5.0	2.0	virginica
## 115	5.8	2.8	5.1	2.4	virginica
## 116	6.4	3.2	5.3	2.3	virginica
## 117	6.5	3.0	5.5	1.8	virginica
## 118	7.7	3.8	6.7	2.2	virginica
## 119	7.7	2.6	6.9	2.3	virginica
## 120	6.0	2.2	5.0	1.5	virginica
## 121	6.9	3.2	5.7	2.3	virginica
## 122	5.6	2.8	4.9	2.0	virginica
## 123	7.7	2.8	6.7	2.0	virginica
## 124	6.3	2.7	4.9	1.8	virginica
## 125	6.7	3.3	5.7	2.1	virginica
## 126	7.2	3.2	6.0	1.8	virginica
## 127	6.2	2.8	4.8	1.8	virginica
## 128	6.1	3.0	4.9	1.8	virginica
## 129	6.4	2.8	5.6	2.1	virginica
## 130	7.2	3.0	5.8	1.6	virginica
## 131	7.4	2.8	6.1	1.9	virginica
## 132	7.9	3.8	6.4	2.0	virginica
## 133	6.4	2.8	5.6	2.2	virginica
## 134	6.3	2.8	5.1	1.5	virginica
## 135	6.1	2.6	5.6	1.4	virginica
## 136	7.7	3.0	6.1	2.3	virginica
## 137	6.3	3.4	5.6	2.4	virginica
## 138	6.4	3.1	5.5	1.8	virginica
## 139	6.0	3.0	4.8	1.8	virginica
## 140	6.9	3.1	5.4	2.1	virginica
## 141	6.7	3.1	5.6	2.4	virginica
## 142	6.9	3.1	5.1	2.3	virginica
## 143	5.8	2.7	5.1	1.9	virginica
## 144	6.8	3.2	5.9	2.3	virginica
## 145	6.7	3.3	5.7	2.5	virginica
## 146	6.7	3.0	5.2	2.3	virginica
## 147	6.3	2.5	5.0	1.9	virginica
## 148	6.5	3.0	5.2	2.0	virginica
## 149	6.2	3.4	5.4	2.3	virginica
## 150	5.9	3.0	5.1	1.8	virginica

2.17.2 More Visualization of the iris dataset

These are basic R functions that allow us to get general information about a dataframe:

- `dim()`
- `head()`
- `View()`
- `class()`
- `str()`

2.17.3 Basic R functions

```
dim(iris)
head(iris)
View(iris)
class(iris)
str(iris)
```

- Select the number of lines to visualize with the head command

```
head(iris, n = 10)
```

```
##      Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1         3.5          1.4          0.2  setosa
## 2           4.9         3.0          1.4          0.2  setosa
## 3           4.7         3.2          1.3          0.2  setosa
## 4           4.6         3.1          1.5          0.2  setosa
## 5           5.0         3.6          1.4          0.2  setosa
## 6           5.4         3.9          1.7          0.4  setosa
## 7           4.6         3.4          1.4          0.3  setosa
## 8           5.0         3.4          1.5          0.2  setosa
## 9           4.4         2.9          1.4          0.2  setosa
## 10          4.9         3.1          1.5          0.1  setosa
```

- Determine the number of columns and rows in the dataframe:

```
dim(iris)
```

```
## [1] 150   5
```

- Determine the `class()` that the dataset belongs to

```
class(iris)
```

```
## [1] "data.frame"
```

2.18 More Visualization

2.18.1 Scatterplot (includes warnings)

```
plot(data=iris, iris$Sepal.Length, iris$Sepal.Width) ## R will complain about this command
```

```
## Warning in plot.window(...): "data" is not a graphical parameter
```

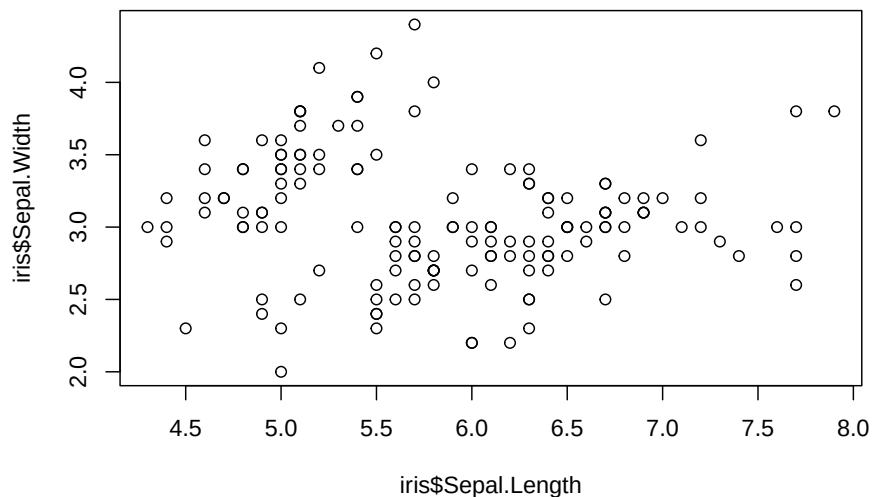
```
## Warning in plot.xy(xy, type, ...): "data" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a  
## graphical parameter
```

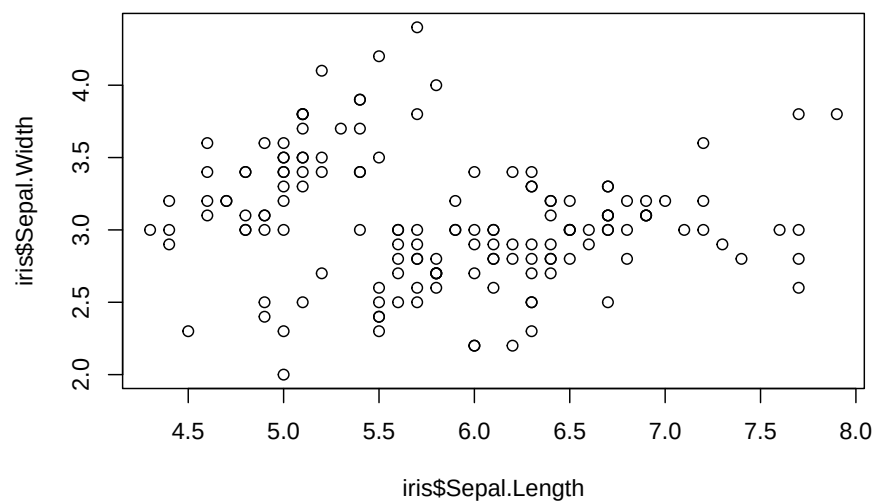
```
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a  
## graphical parameter
```

```
## Warning in box(...): "data" is not a graphical parameter
```

```
## Warning in title(...): "data" is not a graphical parameter
```



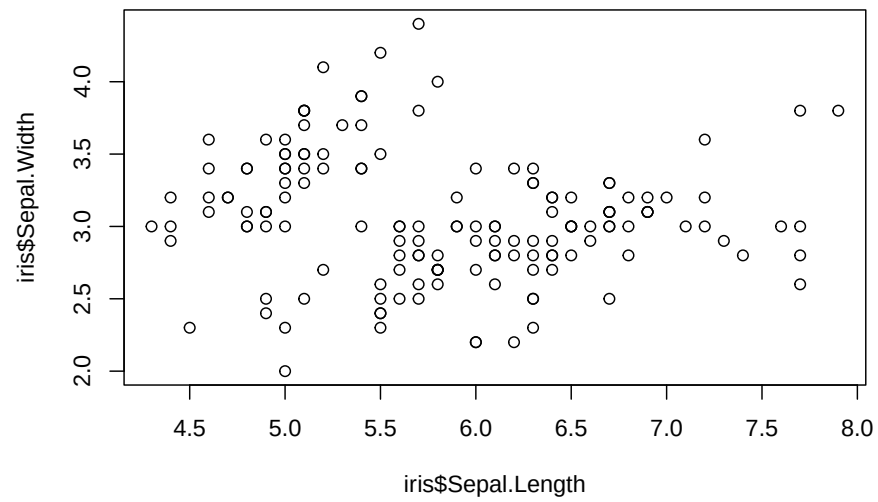
```
plot(iris$Sepal.Length, iris$Sepal.Width) ## According to the error, this was plotted using a dif
```



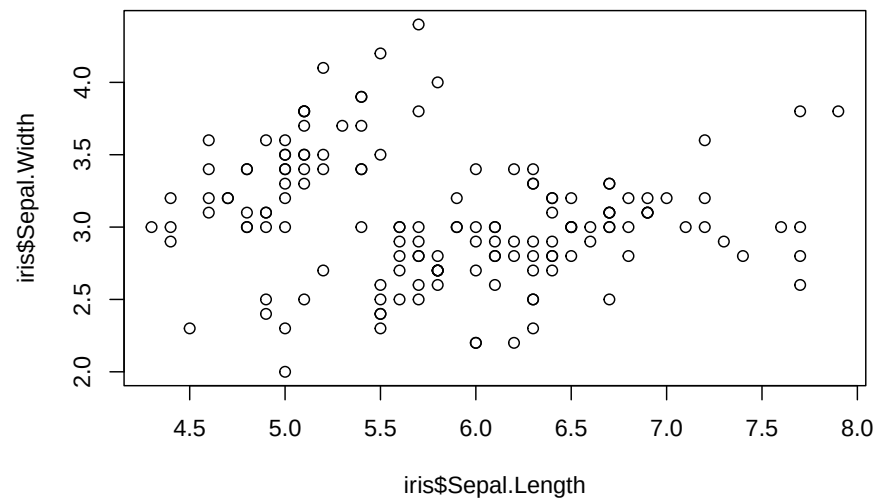
2.18.2 Scatterplot

Plot without warning and message

```
plot(data=iris, iris$Sepal.Length, iris$Sepal.Width) ## R will complain about this command
```

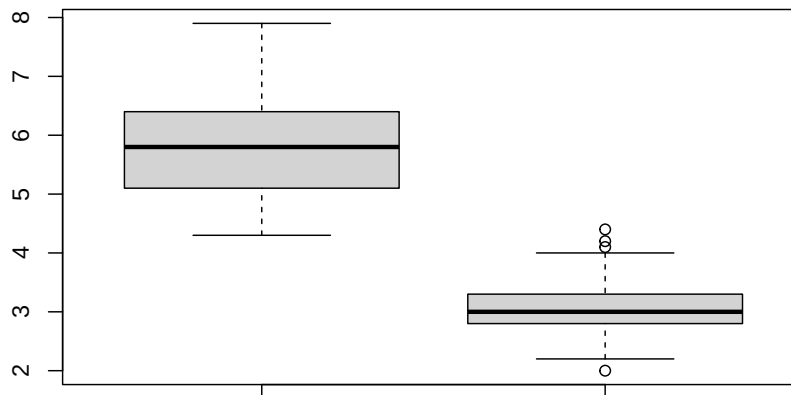


```
plot(iris$Sepal.Length, iris$Sepal.Width) ## According to the error, this was plotted
```



2.18.3 Boxplot

```
boxplot(data=iris, iris$Sepal.Length, iris$Sepal.Width)
```

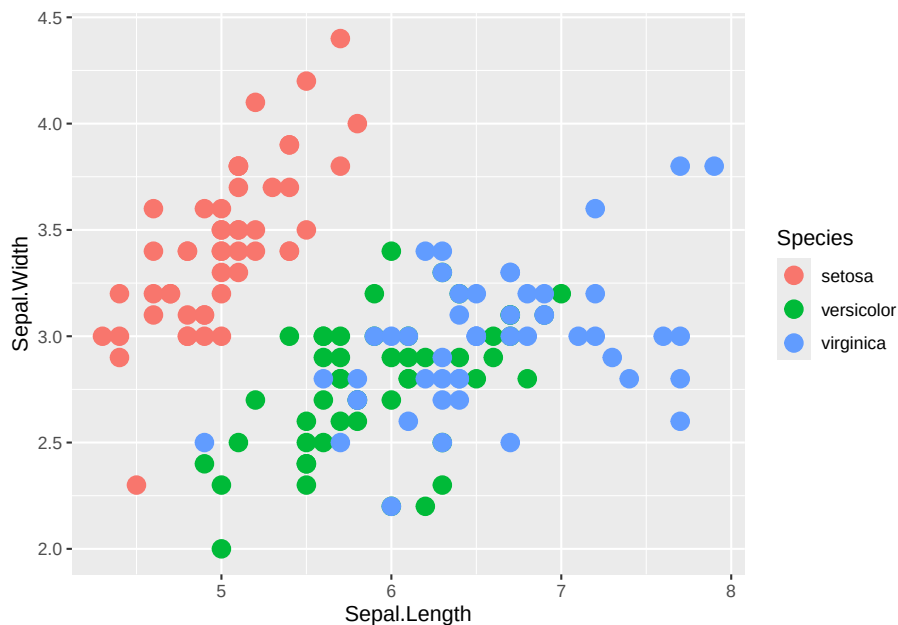


2.19 Data Visualization with specialized libraries

- In R, there are packages designed with the purpose of making good-looking graphics. This is the case with the ggplot2. The chunk below installs the ggplot2 library, loads the library into the R environment and then plots the data present in the iris data-frame.
- You can uncomment the installation line if you need to install it

2.19.1 Ggplot2

```
#install.packages("ggplot2")
library(ggplot2)
ggplot(data=iris, aes(x=Sepal.Length, y=Sepal.Width, color=Species)) + geom_point(size=4)
```



In our activities to visualize human genomic data, we will use a library called qqman, to visualize the biological association, through a plot known as the Manhattan plot.

2.19.2 Visualizing GWAS

- The simplest definition of a GWAS is the statistical or significant association between a phenotype (trait) and a genotype. This association can also be called *biological association*.
- Information about association of SNPs with Huntington's Disease can be found at the Chaves 2019 Huntington's disease paper

2.19.2.1 Package installation

- We need to install and load package qqman

```
## install.packages("qqman")
library(qqman)
```

```
##
```

```
## For example usage please run: vignette('qqman')
```

```
##
```

```
## Citation appreciated but not required:
```

```
## Turner, (2018). qqman: an R package for visualizing GWAS results using Q-Q and manhattan plots
```

```
##
```

2.19.2.2 Assign text file to dataframe object

- After, we load data-frame to be visualized
- Exact location of text file in your system needs to be determined

```
GWAS_TABLE <- read.table("./data/chr4.txt",
                          sep = " ",
                          header = T)
```

2.19.2.3 Get information about object with head()

- Use head() function to inspect the data-frame just loaded

```
head(GWAS_TABLE, n = 10)
```

```
##      CHR      SNP      BP      P
## 1      4 chr4:128096 128096 0.0133500
## 2      4 chr4:516586 516586 0.0076260
## 3      4 chr4:523979 523979 0.0024960
## 4      4 chr4:527217 527217 0.0217400
## 5      4 chr4:566177 566177 0.0008988
## 6      4 chr4:578679 578679 0.0162100
## 7      4 chr4:578790 578790 0.0103700
## 8      4 chr4:579307 579307 0.0334600
## 9      4 chr4:580259 580259 0.0190400
## 10     4 chr4:585318 585318 0.0317600
```

2.19.2.4 Plotting GWAS data-frame

- In this section we use functions plot() and boxplot() to visualize data-frame
- In the y access, we see genomic coordinates and in the x access, p-valoues of the biological association

- Note that depending on the syntax used for plotting, R may complain
- Comments not turned off

```
plot(data=GWAS_TABLE, GWAS_TABLE$BP, GWAS_TABLE$P) ## R complains
```

```
## Warning in plot.window(...): "data" is not a graphical parameter
```

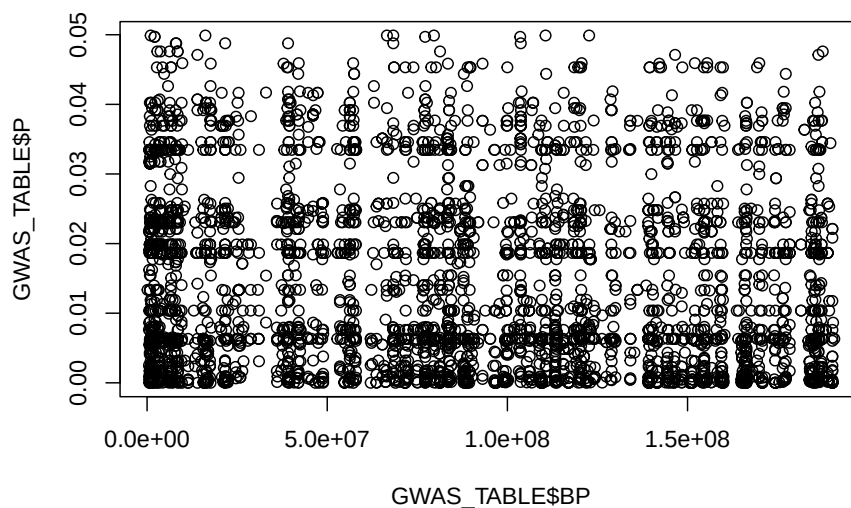
```
## Warning in plot.xy(xy, type, ...): "data" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a  
## graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a  
## graphical parameter
```

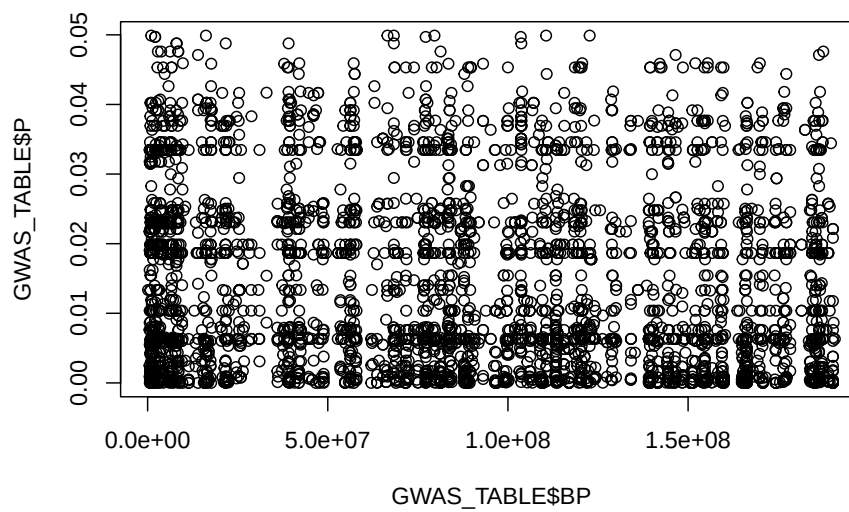
```
## Warning in box(...): "data" is not a graphical parameter
```

```
## Warning in title(...): "data" is not a graphical parameter
```



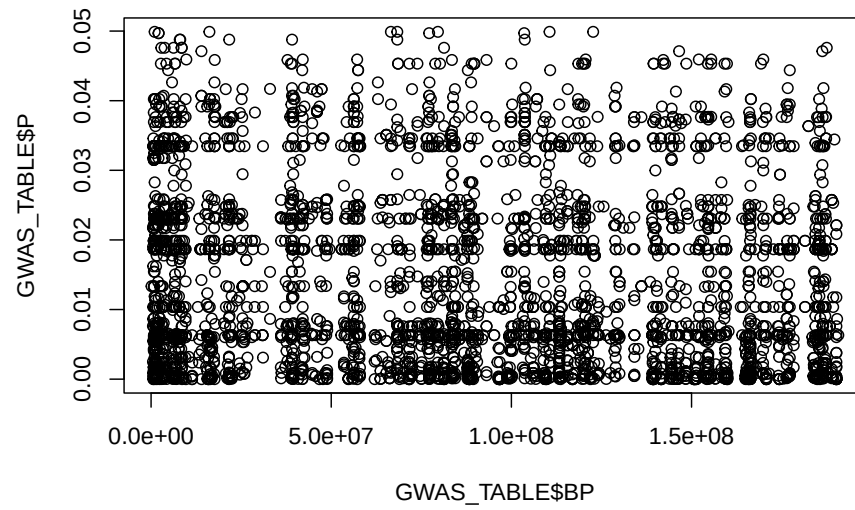
- Turn off comments

```
plot(data=GWAS_TABLE, GWAS_TABLE$BP, GWAS_TABLE$P) ## R complains
```



2.19.2.5 Plotting GWAS data-frame

```
plot(GWAS_TABLE$BP, GWAS_TABLE$P) ## R does not complain
```

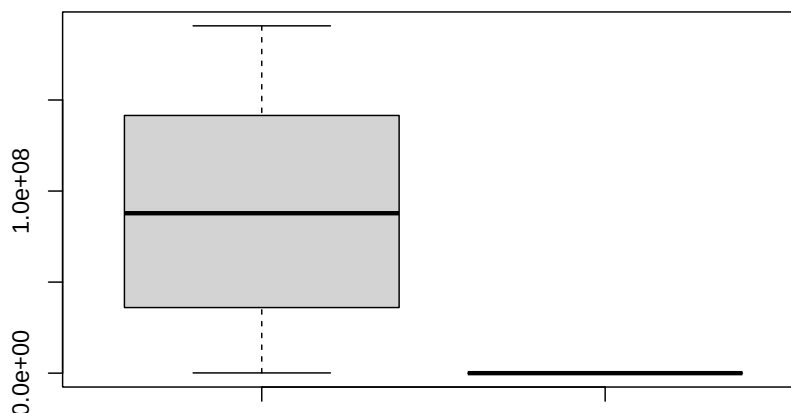


2.19.2.6 Boxplot

- Boxplot

```
boxplot(data=GWAS_TABLE, GWAS_TABLE$BP, GWAS_TABLE$P)
```

2.20. CREATE VECTOR TO HIGHLIGHT GENOME COORDINATES IN MANHATTAN PLOT31



- Compare the chromosomal coordinates and the values of the p-values in the boxplot above
- The chunk below allows one to omit NA values in data-frame

```
GWAS_TABLE_Ommit <- na.omit(GWAS_TABLE)
```

A seguir, criamos uma variável que armazena as posições das SNPs a serem realçadas em verde no Manhattan plot. Estas SNPs são mutações biológica e estatisticamente associadas à Doença de Huntington no gene de uma sortilina, localizada em proximidade física ao gene da proteína huntingtina mutada, a qual é a causadora medeliana (segue as leis de segregação genética de Mendel) da Doença de Huntington.

2.20 Create vector to highlight genome coordinates in Manhattan plot

```
SNP_HIGHLIGHT <- c("chr4:3043512", "chr4:3043513", "chr4:3048207", "chr4:3224216",  
                    "chr4:3231772", "chr4:3233844", "chr4:3235081", "chr4:3235084",  
                    "chr4:3236881", "chr4:3236883", "chr4:3241845", "chr4:3243804",  
                    "chr4:3263138", "chr4:3265130", "chr4:3265710", "chr4:3314646",
```

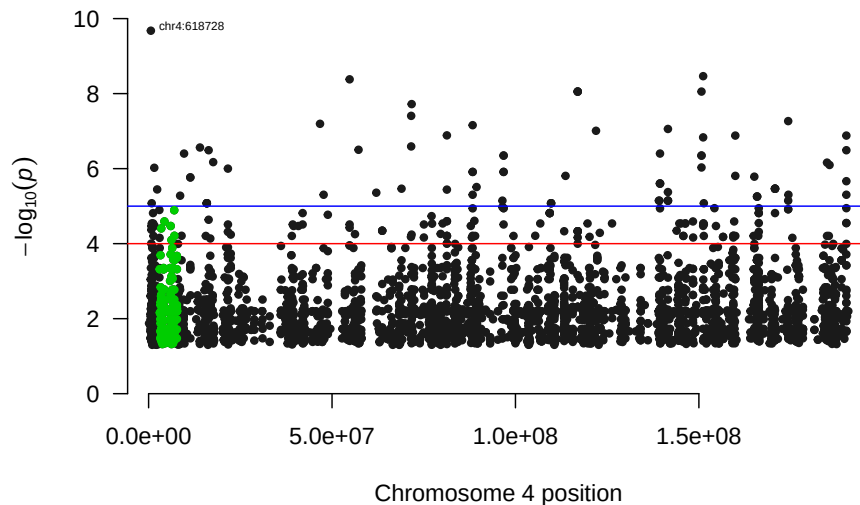
```
"chr4:3380088", "chr4:3409359", "chr4:3411110", "chr4:3415336",  
"chr4:3415378", "chr4:3438643", "chr4:3446091", "chr4:3449886",  
"chr4:3473066", "chr4:3476809", "chr4:3480439", "chr4:3487151",  
"chr4:3496058", "chr4:3496110", "chr4:3506933", "chr4:3508752",  
"chr4:3510957", "chr4:3512690", "chr4:3517746", "chr4:3518190",  
"chr4:3529671", "chr4:3532327", "chr4:3533066", "chr4:3746133",  
"chr4:3747842", "chr4:3748134", "chr4:3765305", "chr4:3765336",  
"chr4:3944253", "chr4:3944752", "chr4:3944888", "chr4:3946166",  
"chr4:3946175", "chr4:3969218", "chr4:4051294", "chr4:4076788",  
"chr4:4103104", "chr4:4103105", "chr4:4109198", "chr4:4109210",  
"chr4:4240627", "chr4:4242705", "chr4:4243668", "chr4:4245210",  
"chr4:4245510", "chr4:4245513", "chr4:4245591", "chr4:4245926",  
"chr4:4245929", "chr4:4246109", "chr4:4246433", "chr4:4246453",  
"chr4:4246457", "chr4:4246497", "chr4:4249414", "chr4:4249415",  
"chr4:4249484", "chr4:4271623", "chr4:4275306", "chr4:4304749",  
"chr4:4318931", "chr4:4318970", "chr4:4319564", "chr4:4319728",  
"chr4:4319750", "chr4:4322078", "chr4:4709657", "chr4:4732282",  
"chr4:4789635", "chr4:4822960", "chr4:4824890", "chr4:4825092",  
"chr4:4825180", "chr4:4865316", "chr4:4865321", "chr4:5018702",  
"chr4:5812778", "chr4:5814082", "chr4:5833660", "chr4:5833899",  
"chr4:5835541", "chr4:5851205", "chr4:5862752", "chr4:5862938",  
"chr4:5862943", "chr4:5901873", "chr4:5905499", "chr4:5906287",  
"chr4:6018891", "chr4:6019046", "chr4:6020190", "chr4:6020367",  
"chr4:6025638", "chr4:6025656", "chr4:6025766", "chr4:6026058",  
"chr4:6083488", "chr4:6204935", "chr4:6235553", "chr4:6237142",  
"chr4:6238466", "chr4:6239906", "chr4:6240929", "chr4:6245022",  
"chr4:6245618", "chr4:6245732", "chr4:6245915", "chr4:6246075",  
"chr4:6246373", "chr4:6246959", "chr4:6290594", "chr4:6292020",  
"chr4:6294095", "chr4:6298375", "chr4:6316092", "chr4:6321396",  
"chr4:6324647", "chr4:6324785", "chr4:6327669", "chr4:6328354",  
"chr4:6328507", "chr4:6333130", "chr4:6333559", "chr4:6333669",  
"chr4:6335966", "chr4:6435341", "chr4:6435486", "chr4:6435926",  
"chr4:6437191", "chr4:6437197", "chr4:6457121", "chr4:6457131",  
"chr4:6457132", "chr4:6568390", "chr4:6570032", "chr4:6570768",  
"chr4:6596360", "chr4:6613252", "chr4:6613462", "chr4:6620991",  
"chr4:6624771", "chr4:6626154", "chr4:6641969", "chr4:6642090",  
"chr4:6644466", "chr4:6644467", "chr4:6644468", "chr4:6647889",  
"chr4:6648300", "chr4:6662665", "chr4:6663319", "chr4:6663715",  
"chr4:6674554", "chr4:6678553", "chr4:6678599", "chr4:6690535",  
"chr4:6698664", "chr4:6698667", "chr4:6698706", "chr4:6720572",  
"chr4:6911679", "chr4:6985889", "chr4:6987394", "chr4:7002344",  
"chr4:7004495", "chr4:7004506", "chr4:7005196", "chr4:7005199",  
"chr4:7024077", "chr4:7024398", "chr4:7029430", "chr4:7031064",  
"chr4:7044357", "chr4:7044380", "chr4:7048842", "chr4:7052115",  
"chr4:7055253", "chr4:7064243", "chr4:7067765", "chr4:7073187",
```

```
"chr4:7074027", "chr4:7677967", "chr4:7701947", "chr4:7702795",  
"chr4:7703505", "chr4:7703807", "chr4:7704795", "chr4:7704818",  
"chr4:7709703", "chr4:7712150", "chr4:7714490", "chr4:7733843",  
"chr4:7735162", "chr4:7735164", "chr4:7736103", "chr4:7736112")
```

2.21 Biological Association Visualization: Manhattan Plot

- Finally, we plot the Manhattan plot graph

```
manhattan(GWAS_TABLE_Ommit,  
          highlight = SNP_HIGHLIGHT,  
          annotateTop = T,  
          annotatePval = 0.20,  
          genomewideline = -log10(0.0001))
```



2.22 References

2.23 Session Info

```
sessionInfo()
```

```
## R version 4.4.1 (2024-06-14)
## Platform: aarch64-apple-darwin20
## Running under: macOS 15.7.4
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Chicago
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] qqman_0.1.9  ggplot2_3.5.1
##
## loaded via a namespace (and not attached):
##  [1] vctrs_0.6.5      cli_3.6.3        knitr_1.48       rlang_1.1.4
##  [5] xfun_0.48        highr_0.11       generics_0.1.3   calibrate_1.7.7
##  [9] labeling_0.4.3   glue_1.8.0       colorspace_2.1-1 htmltools_0.5.8.1
## [13] tinytex_0.53     scales_1.3.0     fansi_1.0.6      rmarkdown_2.28
## [17] grid_4.4.1       evaluate_1.0.0   munsell_0.5.1    tibble_3.2.1
## [21] MASS_7.3-61      fastmap_1.2.0    yaml_2.3.10      lifecycle_1.0.4
## [25] bookdown_0.40    compiler_4.4.1   dplyr_1.1.4      pkgconfig_2.0.3
## [29] rstudioapi_0.16.0 farver_2.1.2     digest_0.6.37    R6_2.5.1
## [33] tidyselect_1.2.1 utf8_1.2.4       pillar_1.9.0     magrittr_2.0.3
## [37] withr_3.0.1      tools_4.4.1      gtable_0.3.5
```

Chapter 3

Processamento de dados públicos de neuroblastoma

3.1 Origem dos dados

Os dados analisados neste módulo têm origem na base de dados genômicos R2.

A plataforma R2 é uma ferramenta de análise e visualização genômica online de livre acesso que pode analisar uma grande coleção de dados públicos, mas também permite a análise protegida de conjuntos de dados privados, conforme indicado por Rijswijk (2024).

A R2 consiste em um banco de dados que armazena as informações genômicas, acoplado a um amplo conjunto de ferramentas para analisar/visualizar os conjuntos de dados.

Mais informações sobre a plataforma R2 pode ser encontradas no site da plataforma:

<https://hgserver1.amc.nl/cgi-bin/r2/main.cgi>

3.2 Carregando dados pre-processados de R2

Nesta parte, vamos carregar uma dataframe contendo dados baixados de R2 em que já fizemos pre-processamento. Vamos gerar algumas visualizações para

nos familiarizarmos com os dados. O objeto `r2_gse62564_GSVA_Metadata.rds` deve ser baixado da pasta no Google Drive no seguinte link: <https://drive.google.com/drive/folders/1Kkos4HK2EP4ntrQOJE0mrHYlpeMa1yi4?usp=sharing>

Uma vez que o arquivo é baixado, pode-se carregá-lo no ambiente R usando a função `readRDS`, como feito abaixo:

```
r2_gse62564_GSVA_Metadata <- readRDS("./data/r2_gse62564_GSVA_Metadata.rds")
```

No neuroblastoma, o principal sistema de classificação de risco classifica os pacientes usando características como idade, estadiamento, histologia e status de amplificação do MYCN.

O prognóstico é a associação entre as características do tumor e a sobrevida do paciente.

O MYCN é um gene que codifica um fator de transcrição muito importante no prognóstico do tumor.

Vamos analisar a expressão do MYCN comparada pelo grupo de risco do neuroblastoma.

Primeiro, vamos ver que tipo de variável é o status do MYCN e a expressão do MYCN.

Verificar a classe da variável `mycn_status`:

```
class(r2_gse62564_GSVA_Metadata$mycn_status)
```

```
## [1] "character"
```

Verificar a classe da variável MYCN:

```
class(r2_gse62564_GSVA_Metadata$MYCN)
```

```
## [1] "character"
```

Ambos são do tipo “character”. Para plotar, precisamos fazer com que a expressão MYCN seja do tipo numérico:

```
r2_gse62564_GSVA_Metadata$MYCN <- as.numeric(
  r2_gse62564_GSVA_Metadata$MYCN)
```

Verificar novamente a classe da variável MYCN:

```
class(r2_gse62564_GSVA_Metadata$MYCN)
```

```
## [1] "numeric"
```

Agora a classe de MYCN é um vetor numérico.

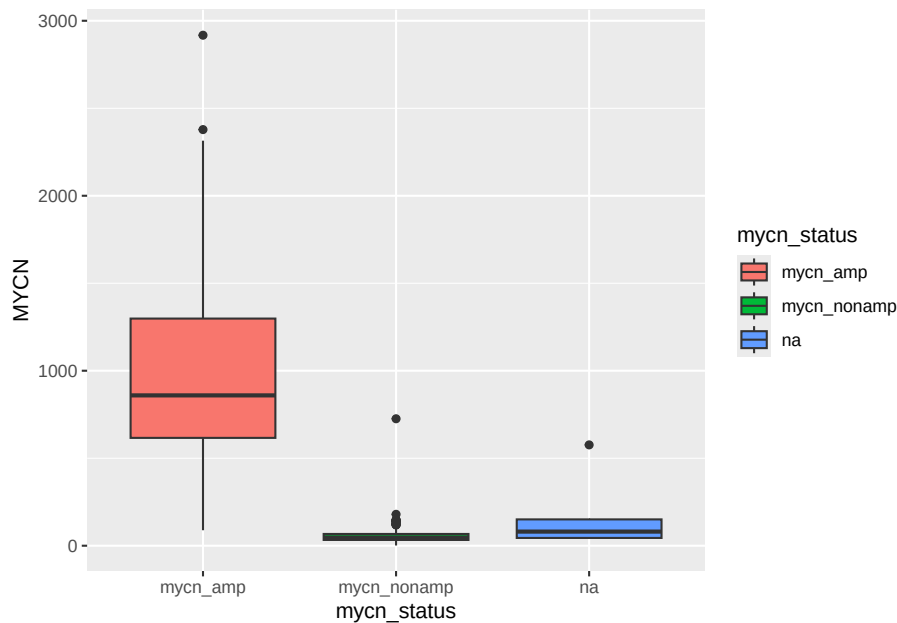
3.3 Primeiros gráficos com dados de neuroblastoma

3.3.1 Mycn_status vs Expressão de MYCN

Vamos plotar uma figura com a variável mycn_status no eixo X e expressão de MYCN no eixo Y.

Vamos tentar ver se o ggplot, que usamos antes, considera a variável mycn_status categórica como ela é agora.

```
library(ggplot2)
ggplot(r2_gse62564_GSVA_Metadata, aes(x=mycn_status, y=MYCN,
                                         fill=mycn_status)) +
  geom_boxplot()
```

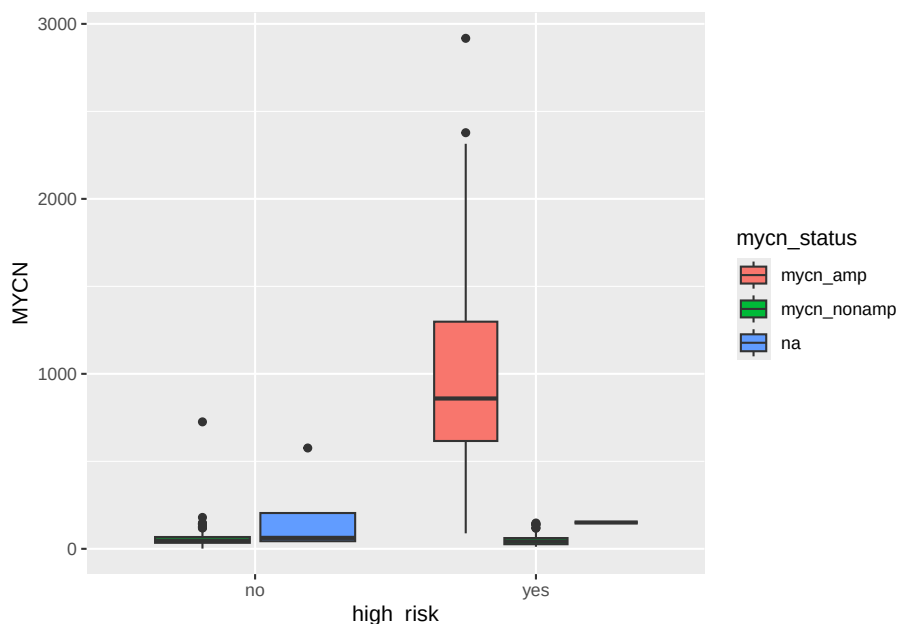


Vemos que existem três categorias de `mycn_status`: `mycn_amp`, `mycn_nonamp` e `na`. `mycn_amp` significa amplificação de MYCN, um fator de mau prognóstico. `mycn_nonamp` significa ausência de amplificação de MYCN. `na` significa que não há informações sobre o status de MYCN. Vemos que a amplificação de MYCN está altamente associada à expressão de MYCN, no eixo Y. Por enquanto, não vamos nos preocupar nem com a formatação dos rótulos X e Y nem com os valores N/A do status de amplificação MYCN.

3.3.2 Alto Risco vs expressão de MYCN

Podemos plotar o status de alto risco vs. amplificação MYCN e deixar o argumento de preenchimento como `mycn_status`

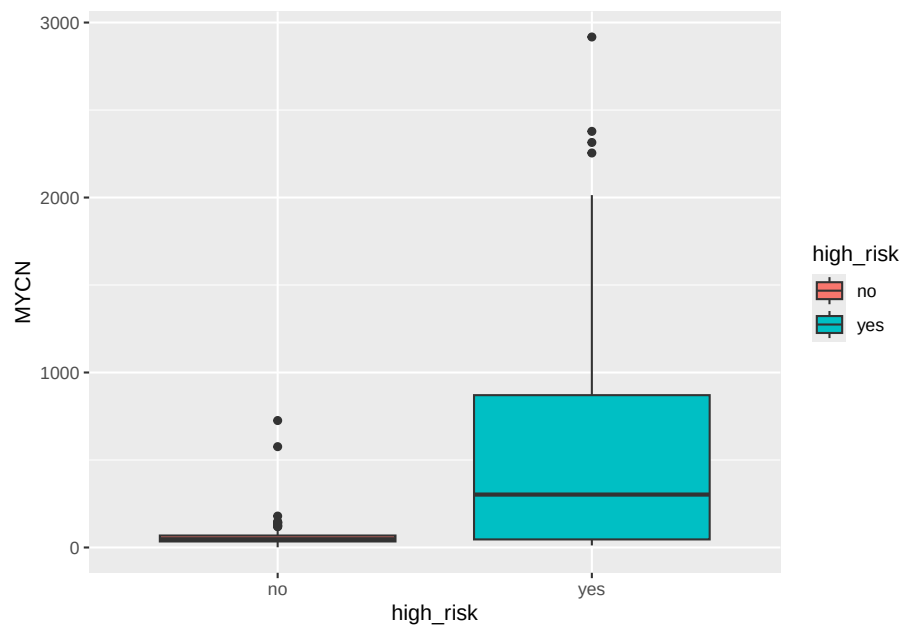
```
library(ggplot2)
ggplot(r2_gse62564_GSVA_Metadata, aes(x=high_risk, y=MYCN,
                                       fill=mycn_status)) +
  geom_boxplot()
```



Aqui vemos que a maior parte dos pacientes de alto risco (High Risk) apresentam também amplificação de MYCN e que nesses pacientes, a expressão de MYCN é alta.

Podemos ainda querer ver apenas a expressão de MYCN no grupo de alto risco, usando o parâmetro `fill=high_risk`:

```
library(ggplot2)
ggplot(r2_gse62564_GSVA_Metadata, aes(x=high_risk, y=MYCN,
                                       fill=high_risk)) +
  geom_boxplot()
```



Onde confirmamos que a expressão de MYCN é maior nos pacientes de alto risco (High Risk).

Chapter 4

Análise de Sobrevivência

Na análise de sobrevivência estudamos como varia a probabilidade de sobrevivência dos pacientes em função do tempo transcorrido desde o início de um estudo. Na atividade proposta para este módulo, vamos avaliar como é a probabilidade de sobrevivência de pacientes de neuroblastoma em função do status de alto risco (se um paciente é de alto risco ou baixo risco). Para dar prosseguimento, carregamos a dataframe de neuroblastoma no ambiente R, como fizemos no módulo anterior:

```
r2_gse62564_GSVA_Metadata <- readRDS( "./data/r2_gse62564_GSVA_Metadata.rds")
```

4.1 Processamento de variáveis para permitir a construção de gráficos

Precisamos definir o status do OS (Overall Survival = sobrevivência geral) como numérico, em vez de um caractere. Fazemos o mesmo para o status EFS (Event Free Survival = sobrevivência livre de evento).

Usamos o método `ifelse` para construir `os_bin_numeric` a partir de `os_bin`, introduzindo números ao invés da palavra evento e não evento. No caso do Overall Survival (OS), a morte do paciente é o evento.

Para usar o comando a seguir, precisamos de usar, antes, uma biblioteca do R chamada `maditr`:

```
library(maditr)
```

```
##  
## To select rows from data: rows(mtcars, am==0)
```

```
##
## Attaching package: 'maditr'

## The following object is masked from 'package:base':
##
##      sort_by
```

Agora podemos usar o operador pipe (`%>%`), que toma o resultado de um comando e passa para o próximo comando:

```
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
  dplyr::mutate(os_bin_numeric =
    ifelse(r2_gse62564_GSVA_Metadata$os_bin == "event", "1", "0"))
```

Agora, fazemos `os_bin_numeric` numérico:

```
r2_gse62564_GSVA_Metadata["os_bin_numeric"] <- as.numeric(r2_gse62564_GSVA_Metadata$os_bin_numeric)
```

Fazemos o mesmo para EFS:

```
## EFS
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
  dplyr::mutate(efs_bin_numeric =
    ifelse(r2_gse62564_GSVA_Metadata$efs_bin == "event", "1", "0"))
```

Agora, fazemos `efs_bin_numeric`, numérico:

```
r2_gse62564_GSVA_Metadata["efs_bin_numeric"] <- as.numeric(r2_gse62564_GSVA_Metadata$efs_bin_numeric)
```

4.2 Fazendo a análise de sobrevivência

4.2.1 Construção da curva de sobrevivência

Usamos a função `survfit` do pacote em R, `survival`, para construir o objeto `fit_os_high_risk`, ajustando o modelo:

```
library(survival)
fit_os_high_risk <- survfit(Surv(as.numeric(os_day), os_bin_numeric) ~ r2_gse62564_GSVA_Metadata$efs_bin_numeric,
  data = r2_gse62564_GSVA_Metadata)
```

Observe que `os_day` é uma variável contendo o número de dias até o evento de interesse.

4.2.2 Plotagem da curva de sobrevivência

Usamos a função `ggsurvplot` do pacote em R `survminer` para plotar a curva de sobrevivência:

```
library(survminer)
```

```
## Loading required package: ggpubr
```

```
##
```

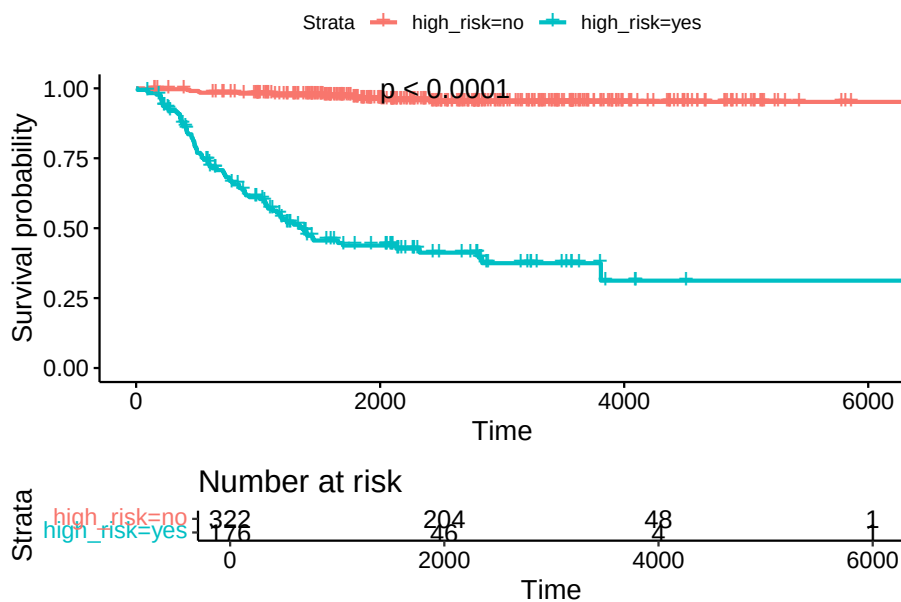
```
## Attaching package: 'survminer'
```

```
## The following object is masked from 'package:survival':
```

```
##
```

```
##      myeloma
```

```
ggsurvplot(fit_os_high_risk,
  data = r2_gse62564_GSVA_Metadata,
  risk.table = TRUE,
  # risk.table = "abs_pct",
  pval = TRUE,
  pval.coord = c(2000, 1),
)
```



4.2.3 Hipóxia no microambiente tumoral

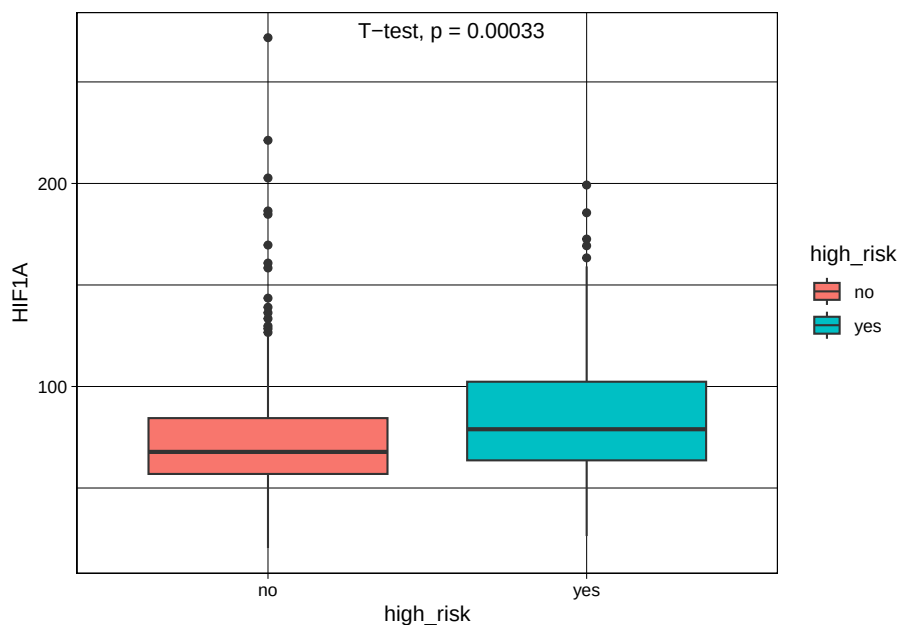
Vamos agora procurar estabelecer uma relação entre hipóxia e neuroblastoma de alto risco. Chama-se hipóxia à condição de baixa concentração de oxigênio que se estabelece no microambiente do tumor à medida que o câncer progride.

HIF1A é um fator de transcrição sensível aos níveis de oxigênio. Precisamos tornar a coluna HIF1A (variável), numérica, como fizemos com MYCN acima:

```
r2_gse62564_GSVA_Metadata$HIF1A <- as.numeric(r2_gse62564_GSVA_Metadata$HIF1A)
```

Compare as médias de HIF1A entre os grupos de alto risco usando ggplot2:

```
ggplot(r2_gse62564_GSVA_Metadata, aes(x=high_risk, y=HIF1A, fill=high_risk)) +  
  geom_boxplot() +  
  stat_compare_means(method = "t.test", label.x = 1.4) +  
  theme_linedraw()
```



Vemos que para este conjunto de dados, a expressão de HIF1A é maior no grupo de alto risco, o que está de acordo com a noção de que aumento da hipóxia leva a maior malignidade na progressão tumoral.

4.2.4 Análise de sobrevivência do gene da hipóxia

Como vemos que indivíduos de alto risco apresentam maior expressão de HIF1A, vamos analisar também a análise de sobrevivência de HIF1A. Para isso vamos separar em dois grupos de pacientes, baseados na expressão do HIF1A. Construindo a variável HIF1A em dois quantis, um com alta expressão de HIF1A e um com baixa expressão de HIF1A. Para isso vamos usar a função `ntile()` do pacote `tidyverse`:

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v lubridate  1.9.3      v tibble     3.2.1
## v purrr      1.0.2      v tidyr      1.3.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::between()      masks maditr::between()
## x dplyr::coalesce()     masks maditr::coalesce()
## x readr::cols()         masks maditr::cols()
## x dplyr::filter()       masks stats::filter()
## x dplyr::first()        masks maditr::first()
## x dplyr::lag()          masks stats::lag()
## x dplyr::last()         masks maditr::last()
## x purrr::transpose()    masks maditr::transpose()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
  mutate(quantile = ntile(HIF1A, 2))
```

Fazendo da variável quantil HIF1A um fator

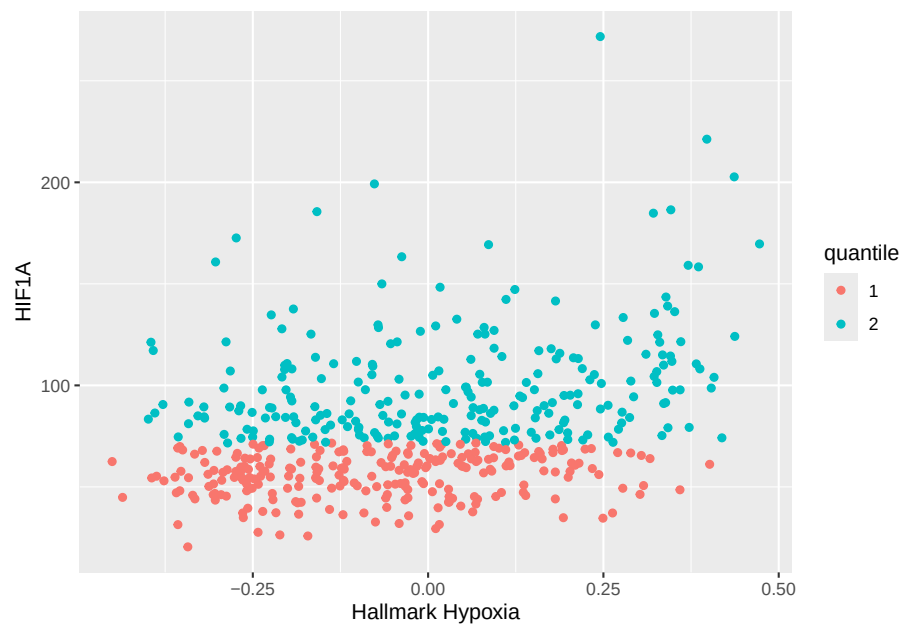
```
r2_gse62564_GSVA_Metadata$quantile <- as.factor(r2_gse62564_GSVA_Metadata$quantile )
```

Gráfico de dispersão HALLMARK HYPOXIA vs. HIF1A

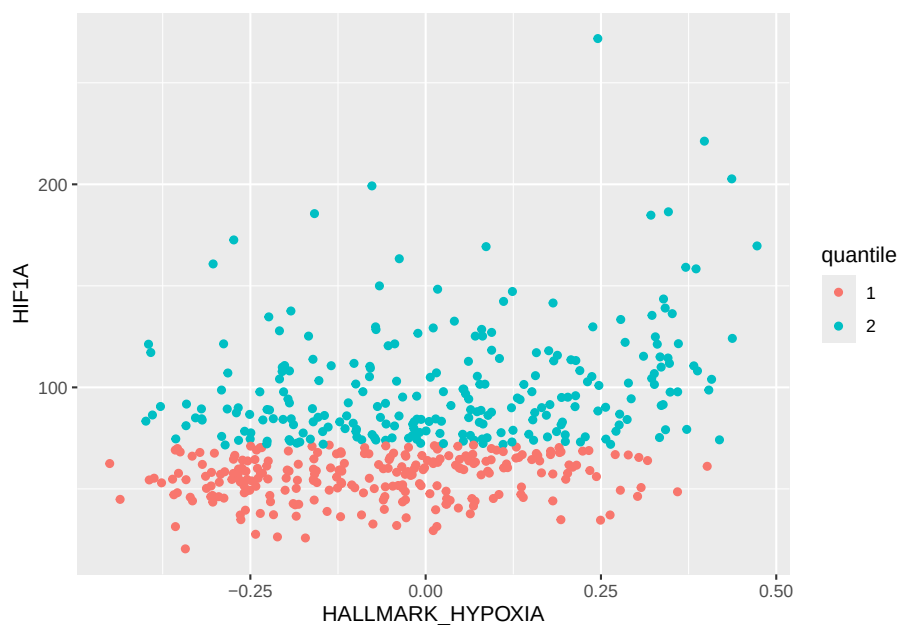
```
r2_gse62564_GSVA_Metadata$HALLMARK_HYPOXIA <- as.numeric(r2_gse62564_GSVA_Metadata$HALLMARK_HYPOXIA)

qplot(HALLMARK_HYPOXIA, HIF1A,
      data = r2_gse62564_GSVA_Metadata,
      colour=quantile,
      ylab = "HIF1A",
      xlab = "Hallmark Hypoxia")
```

```
## Warning: `qplot()` was deprecated in ggplot2 3.4.0.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



```
library(ggplot2)  
# Basic scatter plot  
ggplot(r2_gse62564_GSVA_Metadata, aes(x=HALLMARK_HYPOXIA, y=HIF1A, color=quantile)) +
```



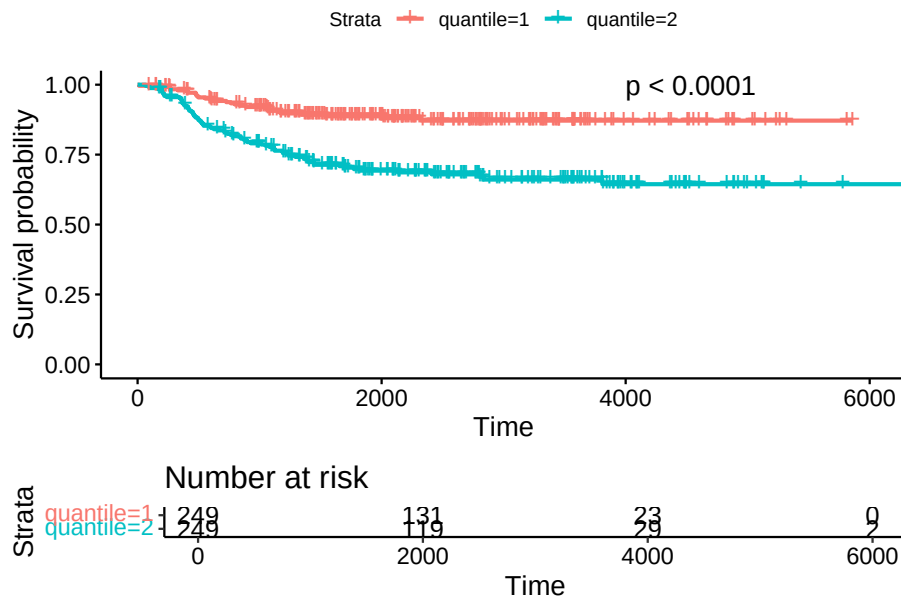
Vemos que o quantil 2 tem valores de expressão de HIF1A maiores que os valores do quantil 1.

Construímos o objeto de sobrevivência ajustado com a variável quantil HIF1A:

```
library(survival)
fit_os_HIF1A <- survfit(Surv(as.numeric(os_day), os_bin_numeric) ~ r2_gse62564_GSVA_Metadata$quantile,
  data = r2_gse62564_GSVA_Metadata)
```

E finalmente traçamos a curva de sobrevivência do gráfico do Overall Survival:

```
library(survminer)
ggsurvplot(fit_os_HIF1A,
  data = r2_gse62564_GSVA_Metadata,
  risk.table = TRUE,
  # risk.table = "abs_pct",
  pval = TRUE,
  pval.coord = c(4000, 1),
)
```



Observamos que os maiores valores na expressão de HIF1A (segundo quantil ou quantil 2. No inglês é second quantile) levam a menor probabilidade de sobrevivência. Vemos que no dataset, o segundo quantil tem menores probabilidades de sobrevivência.

4.2.4.1 Event-free survival

Construímos a variável quantil HIF1A:

```
library(tidyverse)
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
  mutate(quantile = ntile(HIF1A, 2))
```

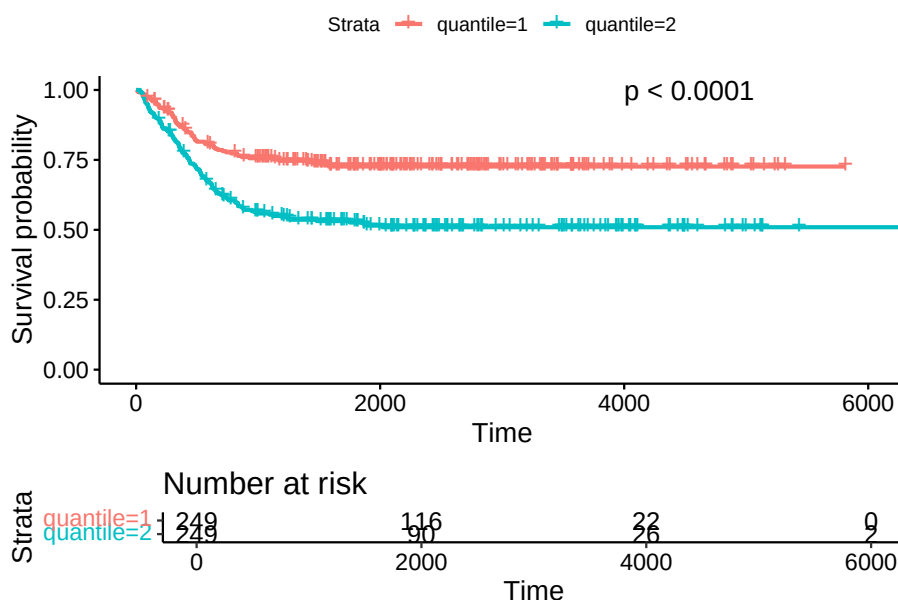
Construímos objeto de sobrevivência EFS, `fit_efs_HIF1A`

```
library(survival)
fit_efs_HIF1A <- survfit(Surv(as.numeric(efs_day), efs_bin_numeric) ~ r2_gse62564_GSVA_Metadata,
  data = r2_gse62564_GSVA_Metadata)
```

Curva de sobrevivência do gráfico EFS, usando a função `ggsurvplot`

```
library(survminer)
ggsurvplot(fit_efs_HIF1A,
```

```
data = r2_gse62564_GSVA_Metadata,
risk.table = TRUE,
# risk.table = "abs_pct",
pval = TRUE,
pval.coord = c(4000, 1),
)
```



4.2.5 Relação do HIF1A com a idade no diagnóstico

Precisamos criar o objeto `r2_gse62564_GSVA_Metadata_Spaces_survival`

A função `compare_means` não consegue lidar com espaços no nome da variável de exaustão de células T.

Teremos que usar um truque para lidar com os espaços em “T cell exhaustion”.

Usaremos o pacote `stringr` para inserir pontos (.) substituindo os espaços no dataframe `r2_gse62564_GSVA_Metadata`.

```
r2_gse62564_GSVA_Metadata_Spaces <- r2_gse62564_GSVA_Metadata
library(stringr)
names(r2_gse62564_GSVA_Metadata_Spaces) <- str_replace_all(names(r2_gse62564_GSVA_Metadata_Spaces),
```

Criando o objeto `r2_gse62564_GSVA_Metadata_Spaces_survival`:

```
r2_gse62564_GSVA_Metadata_Spaces_survival <- r2_gse62564_GSVA_Metadata_Spaces
```

Fazendo age_at_diagnosis numérico para plotar os quantis

```
r2_gse62564_GSVA_Metadata_Spaces_survival$age_at_diagnosis <- as.numeric(r2_gse62564_GSVA_Metadata_Spaces_survival$age_at_diagnosis)
```

Verificando a idade máxima no diagnóstico

```
max(r2_gse62564_GSVA_Metadata_Spaces_survival$age_at_diagnosis)
```

```
## [1] 8983
```

```
min(r2_gse62564_GSVA_Metadata_Spaces_survival$age_at_diagnosis)
```

```
## [1] 0
```

Construir quantis para a idade no diagnóstico (age_qtls)

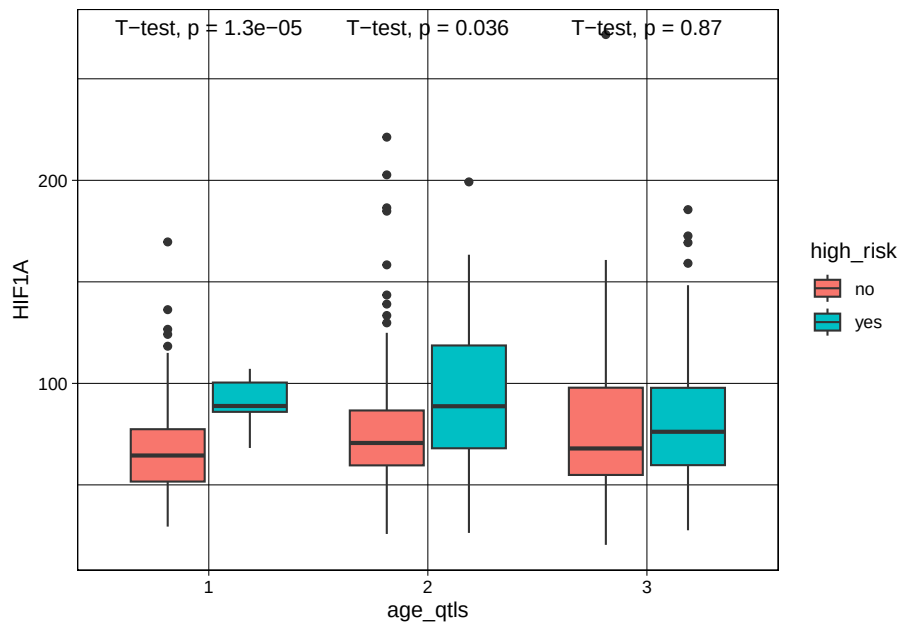
```
r2_gse62564_GSVA_Metadata_Spaces_survival_age <- r2_gse62564_GSVA_Metadata_Spaces_survival %>%
  mutate(age_qtls = ntile(age_at_diagnosis, 3))
```

Faça age_qtls como fator

```
r2_gse62564_GSVA_Metadata_Spaces_survival_age$age_qtls <- as.factor(r2_gse62564_GSVA_Metadata_Spaces_survival_age$age_qtls)
```

Gráfico age_qtls vs. HIF1A

```
ggplot(r2_gse62564_GSVA_Metadata_Spaces_survival_age, aes(x=age_qtls, y=HIF1A, fill=age_qtls)) +
  geom_boxplot() +
  stat_compare_means(method = "t.test") +
  theme_linedraw()
```



Chapter 5

Accessing the Raça/Cor Variable in DATASUS

5.1 Load libraries

```
library(microdatasus)
library(dplyr)
library(tidyverse)
library(maditr)
library(kableExtra)
```

5.2 Constrir dataframe (df) com identificação racial

```
racacor_df <- readRDS("./data/racacor_df.rds")
racacor_df
```

Transforma coluna com caracteres em nomes de raça/cor

```
racacor_df <- racacor_df %>%  
  dplyr::mutate(raca_cor_factor = ifelse(racacor_df$RACACOR == "1", "branca",  
                                         ifelse(racacor_df$RACACOR == "2", "preta",  
                                                  ifelse(racacor_df$RACACOR == "3", "amarela",  
                                                         ifelse(racacor_df$RACACOR == "4", "indigena",  
                                                                ifelse(racacor_df$RACACOR == "5", "outros",  
                                                                                           ifelse(racacor_df$RACACOR == "6", "nao declarada",
```

```
racacor_df
```

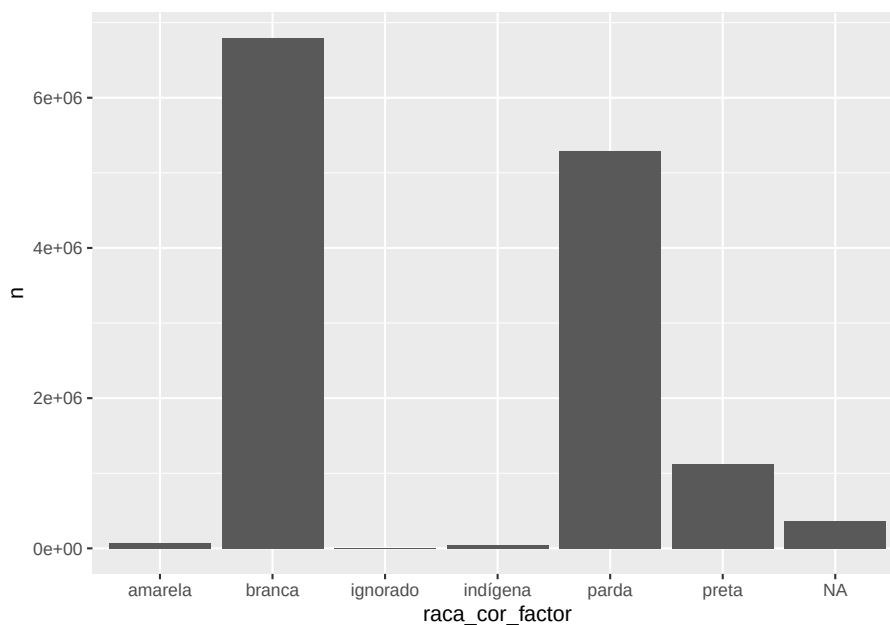
```
##      RACACOR      n  raca_cor_factor
## 1      1 6794687      branca
## 2      4 5295315      parda
## 3      2 1127508      preta
## 4    <NA> 364106      <NA>
## 5      3  74842      amarela
## 6      5  46782      indígena
## 7      9      5      ignorado
```

5.4. GRÁFICO DE BARRAS POR CATEGORIA DA VARIÁVEL RAÇA/COR⁵⁵

5.4 Gráfico de barras por categoria da Variável Raça/Cor

Nesta parte, poderemos estimar o número de indivíduos sob risco, considerando as categorias raciais brasileiras.

```
library(ggplot2)
# Basic barplot
ggplot(data=racacor_df, aes(x=raca_cor_factor, y=n)) +
  geom_bar(stat="identity")
```



```
dados_agregados_year <- readRDS("./data/dados_agregados_year.rds")
dados_agregados_year
```

```
## # A tibble: 10 x 2
##   year      n
##   <chr>  <int>
## 1 2015 1264175
```

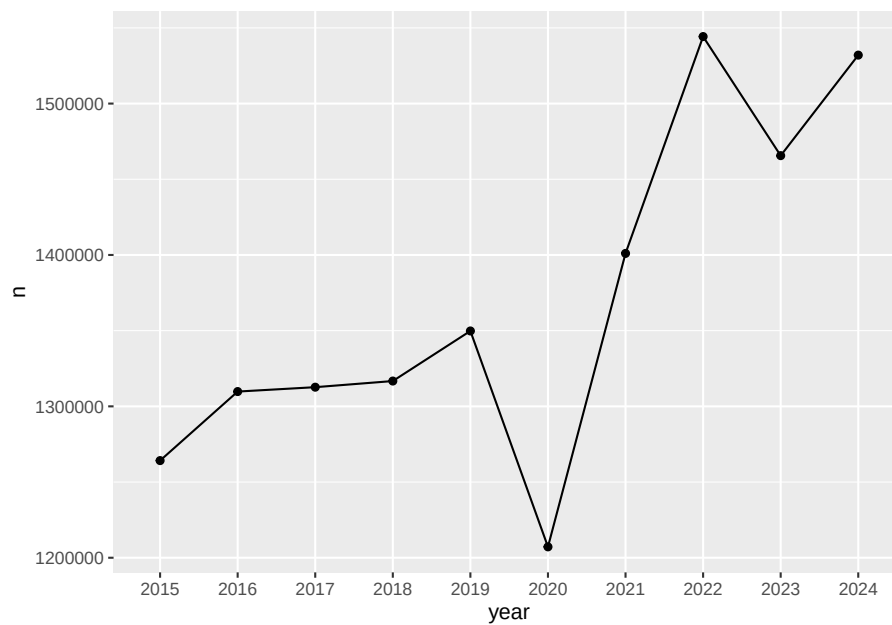
```
## 2 2016 1309774
## 3 2017 1312663
## 4 2018 1316719
## 5 2019 1349801
## 6 2020 1207189
## 7 2021 1401033
## 8 2022 1544266
## 9 2023 1465610
## 10 2024 1532015
```

Remover NAs dos dados agregados por ano

```
dados_agregados_year <- dados_agregados_year[ which(
  dados_agregados_year$year != "NA"), ]
```

5.5 Mortalidade Anual Total

```
library(ggplot2)
# Basic line plot with points
ggplot(data=dados_agregados_year, aes(x=year, y=n, group=1)) +
  geom_line() +
  geom_point()
```



5.5.1 Mortalidade Anual Total Usando Variável Raça/Cor

Mortalidade anual total (figura acima) e mortalidade anual usando a variável Raça/Cor (figura abaixo) foram feitas usando ? como referência.

Carregar dados_appended_melted

```
dados_appended_melted <- readRDS("./data/dados_appended_melted.rds")
dados_appended_melted
```

```
##   year variable  value
## 1  2015         1 643928
## 2  2016         1 665737
## 3  2017         1 663147
## 4  2018         1 668707
## 5  2019         1 686474
```

## 6	2020	1	523848
## 7	2021	1	644210
## 8	2022	1	787363
## 9	2023	1	737590
## 10	2024	1	773683
## 11	2015	2	93830
## 12	2016	2	97747
## 13	2017	2	99745
## 14	2018	2	102889
## 15	2019	2	107306
## 16	2020	2	107171
## 17	2021	2	122852
## 18	2022	2	131005
## 19	2023	2	129047
## 20	2024	2	135916
## 21	2015	3	7051
## 22	2016	3	7276
## 23	2017	3	7375
## 24	2018	3	7524
## 25	2019	3	7359
## 26	2020	3	4716
## 27	2021	3	5313
## 28	2022	3	9261
## 29	2023	3	9053
## 30	2024	3	9914
## 31	2015	4	458807
## 32	2016	4	483026
## 33	2017	4	497839
## 34	2018	4	496106
## 35	2019	4	509936
## 36	2020	4	531808
## 37	2021	4	587323
## 38	2022	4	582469
## 39	2023	4	561438
## 40	2024	4	586563
## 41	2015	5	3640
## 42	2016	5	3867
## 43	2017	5	3911
## 44	2018	5	4261
## 45	2019	5	4273
## 46	2020	5	5173
## 47	2021	5	5361
## 48	2022	5	5343
## 49	2023	5	5398
## 50	2024	5	5555
## 51	2015	9	0

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## 10 2024	1 773683	branca
## 11 2015	2 93830	preta
## 12 2016	2 97747	preta
## 13 2017	2 99745	preta
## 14 2018	2 102889	preta
## 15 2019	2 107306	preta
## 16 2020	2 107171	preta
## 17 2021	2 122852	preta
## 18 2022	2 131005	preta
## 19 2023	2 129047	preta
## 20 2024	2 135916	preta
## 21 2015	3 7051	amarela
## 22 2016	3 7276	amarela
## 23 2017	3 7375	amarela
## 24 2018	3 7524	amarela
## 25 2019	3 7359	amarela
## 26 2020	3 4716	amarela
## 27 2021	3 5313	amarela
## 28 2022	3 9261	amarela
## 29 2023	3 9053	amarela
## 30 2024	3 9914	amarela
## 31 2015	4 458807	parda
## 32 2016	4 483026	parda
## 33 2017	4 497839	parda
## 34 2018	4 496106	parda
## 35 2019	4 509936	parda
## 36 2020	4 531808	parda
## 37 2021	4 587323	parda
## 38 2022	4 582469	parda
## 39 2023	4 561438	parda
## 40 2024	4 586563	parda
## 41 2015	5 3640	indígena
## 42 2016	5 3867	indígena
## 43 2017	5 3911	indígena
## 44 2018	5 4261	indígena
## 45 2019	5 4273	indígena
## 46 2020	5 5173	indígena
## 47 2021	5 5361	indígena
## 48 2022	5 5343	indígena
## 49 2023	5 5398	indígena
## 50 2024	5 5555	indígena
## 51 2015	9 0	ignorado
## 52 2016	9 5	ignorado
## 53 2017	9 0	ignorado
## 54 2018	9 0	ignorado
## 55 2019	9 0	ignorado

```
## 56 2020      9      0      ignorado
## 57 2021      9      0      ignorado
## 58 2022      9      0      ignorado
## 59 2023      9      0      ignorado
## 60 2024      9      0      ignorado
## 61 2015     NA 56919          0
## 62 2016     NA 52116          0
## 63 2017     NA 40646          0
## 64 2018     NA 37232          0
## 65 2019     NA 34453          0
## 66 2020     NA 34473          0
## 67 2021     NA 35974          0
## 68 2022     NA 28825          0
## 69 2023     NA 23084          0
## 70 2024     NA 20384          0
```

Remove NA in year

```
dados_appended_melted <- dados_appended_melted[ which(
  dados_appended_melted$year != "NA"), ]
```

Remove 0 in raça/cor

```
dados_appended_melted <- dados_appended_melted[ which(
  dados_appended_melted$raca_cor_factor != "0"), ]
```

```
dados_appended_melted
```

```
##   year variable  value raca_cor_factor
## 1  2015      1 643928      branca
## 2  2016      1 665737      branca
## 3  2017      1 663147      branca
## 4  2018      1 668707      branca
## 5  2019      1 686474      branca
## 6  2020      1 523848      branca
## 7  2021      1 644210      branca
## 8  2022      1 787363      branca
## 9  2023      1 737590      branca
## 10 2024      1 773683      branca
```

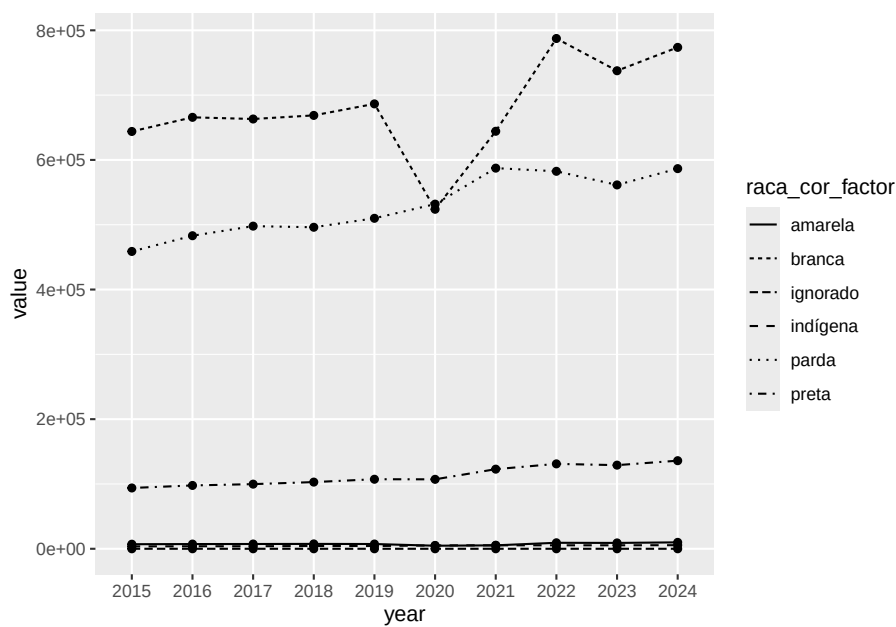
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## 11 2015	2	93830	preta
## 12 2016	2	97747	preta
## 13 2017	2	99745	preta
## 14 2018	2	102889	preta
## 15 2019	2	107306	preta
## 16 2020	2	107171	preta
## 17 2021	2	122852	preta
## 18 2022	2	131005	preta
## 19 2023	2	129047	preta
## 20 2024	2	135916	preta
## 21 2015	3	7051	amarela
## 22 2016	3	7276	amarela
## 23 2017	3	7375	amarela
## 24 2018	3	7524	amarela
## 25 2019	3	7359	amarela
## 26 2020	3	4716	amarela
## 27 2021	3	5313	amarela
## 28 2022	3	9261	amarela
## 29 2023	3	9053	amarela
## 30 2024	3	9914	amarela
## 31 2015	4	458807	parda
## 32 2016	4	483026	parda
## 33 2017	4	497839	parda
## 34 2018	4	496106	parda
## 35 2019	4	509936	parda
## 36 2020	4	531808	parda
## 37 2021	4	587323	parda
## 38 2022	4	582469	parda
## 39 2023	4	561438	parda
## 40 2024	4	586563	parda
## 41 2015	5	3640	indígena
## 42 2016	5	3867	indígena
## 43 2017	5	3911	indígena
## 44 2018	5	4261	indígena
## 45 2019	5	4273	indígena
## 46 2020	5	5173	indígena
## 47 2021	5	5361	indígena
## 48 2022	5	5343	indígena
## 49 2023	5	5398	indígena
## 50 2024	5	5555	indígena
## 51 2015	9	0	ignorado
## 52 2016	9	5	ignorado
## 53 2017	9	0	ignorado
## 54 2018	9	0	ignorado
## 55 2019	9	0	ignorado
## 56 2020	9	0	ignorado

```
## 57 2021      9      0      ignorado
## 58 2022      9      0      ignorado
## 59 2023      9      0      ignorado
## 60 2024      9      0      ignorado
```

Visualizar plot

```
ggplot(dados_appended_melted, aes(x=year, y=value, group=raca_cor_factor)) +
  geom_line(aes(linetype=raca_cor_factor)) +
  geom_point()
```



```
# wait to change the variable names:
```

```
# names(dados_appended_melted)[names(dados_appended_melted) == 'variable'] <- 'RACACOR'
# names(dados_appended_melted)[names(dados_appended_melted) == 'value'] <- 'n'
```

5.6 Mortalidade Anual Total por Neuroblastoma

```
dados_nb <- readRDS("./data/dados_nb.rds")
```

```
# Checar dados_nb
head(dados_nb)
```

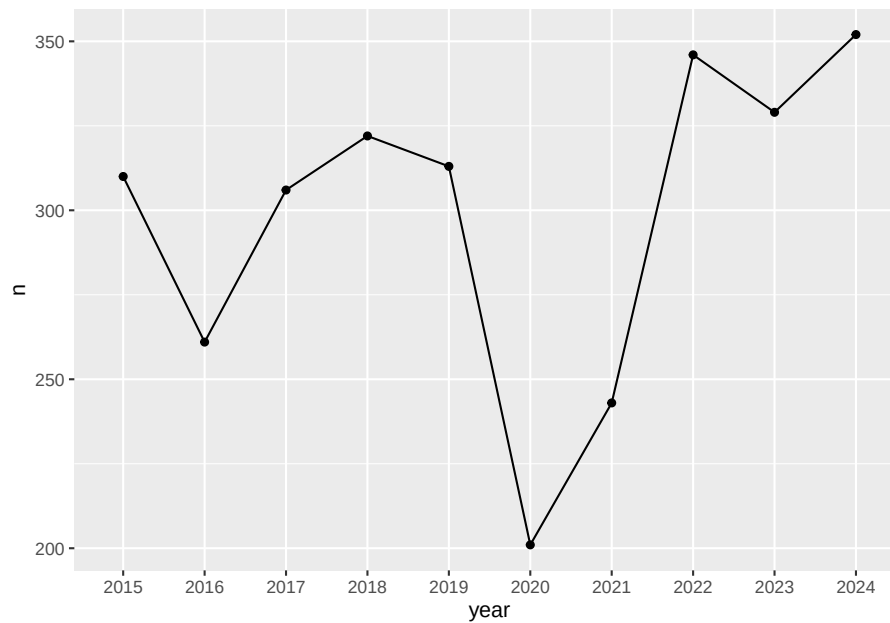
```
##          CONTADOR RACACOR  DTOBITO CAUSABAS  DTNASC  IDADE      YMD year
## 876...17551      876        4 20012016    C749 10122010   405 2016-01-20 2016
## 5811...56566   562030      1 14052018    C749 22042018   222 2018-05-14 2018
## 8831...77296   613331      1 11062019    C749 31011964   455 2019-06-11 2019
## 14106...82571  987213      4 01042019    C749 12021950   469 2019-04-01 2019
## 900...87692    78999      1 08032020    C749 02042014   405 2020-03-08 2020
## 8259...95051   470277      1 15022020    C749 21052005   414 2020-02-15 2020
```

```
dados_agregados_nb_year <- dados_nb %>%
  group_by(year) %>%
  summarise(n = n())

# Checar dados_agregados_nb_year
dados_agregados_nb_year
```

```
## # A tibble: 10 x 2
##   year      n
##   <chr> <int>
## 1 2015    310
## 2 2016    261
## 3 2017    306
## 4 2018    322
## 5 2019    313
## 6 2020    201
## 7 2021    243
## 8 2022    346
## 9 2023    329
## 10 2024    352
```

```
ggplot(data=dados_agregados_nb_year, aes(x=year, y=n, group=1)) +
  geom_line() +
  geom_point()
```



```
dados_nb_race_year_selcd <- dados_nb %>%
  select(RACACOR, year)
```

```
# Checar dados_nb_race_year_selcd
head(dados_nb_race_year_selcd)
```

```
##           RACACOR year
## 876...17551      4 2016
## 5811...56566     1 2018
## 8831...77296     1 2019
## 14106...82571    4 2019
## 900...87692      1 2020
## 8259...95051     1 2020
```

```
# dados_nb_race_2013_selcd <- dados_estados_nb_2013 %>%
#   select(RACACOR, CAUSABAS)
```

use `cast()` to convert data frame from long to wide format

```
library(reshape2)
```

```
##
## Attaching package: 'reshape2'
```

```
## The following object is masked from 'package:tidyr':
##
##      smiths
```

```
## The following objects are masked from 'package:maditr':
##
##      dcast, melt
```

```
wide_dados_nb_appended <- dcast(dados_nb_race_year_selcd, year ~ RACACOR)
```

```
## Using year as value column: use value.var to override.
```

```
## Aggregation function missing: defaulting to length
```

Verify wide_dados_nb_appended

```
wide_dados_nb_appended
```

```
##      year    1  2 3    4 5 NA
## 1  2015 175 14 1 104 1 15
## 2  2016 146 15 1  86 0 13
## 3  2017 178 16 1 101 0 10
## 4  2018 201 17 1  88 1 14
## 5  2019 159 18 1 121 0 14
## 6  2020 113 13 1  66 0  8
## 7  2021 121 20 2  93 0  7
## 8  2022 187 24 0 129 0  6
## 9  2023 177 20 0 121 2  9
## 10 2024 199 15 2 127 1  8
```

Melt the data frame

```
dados_nb_appended_melted <- melt(wide_dados_nb_appended, id.vars = "year")
```

Verificar dados_nb_appended_melted

```
dados_nb_appended_melted
```

```
##   year variable value
## 1 2015         1   175
## 2 2016         1   146
## 3 2017         1   178
## 4 2018         1   201
## 5 2019         1   159
## 6 2020         1   113
## 7 2021         1   121
## 8 2022         1   187
## 9 2023         1   177
## 10 2024        1   199
## 11 2015         2    14
## 12 2016         2    15
## 13 2017         2    16
## 14 2018         2    17
## 15 2019         2    18
## 16 2020         2    13
## 17 2021         2    20
## 18 2022         2    24
## 19 2023         2    20
## 20 2024         2    15
## 21 2015         3     1
## 22 2016         3     1
## 23 2017         3     1
## 24 2018         3     1
## 25 2019         3     1
## 26 2020         3     1
## 27 2021         3     2
## 28 2022         3     0
## 29 2023         3     0
## 30 2024         3     2
## 31 2015         4   104
## 32 2016         4    86
## 33 2017         4   101
## 34 2018         4    88
## 35 2019         4   121
```

```
## 36 2020      4    66
## 37 2021      4    93
## 38 2022      4   129
## 39 2023      4   121
## 40 2024      4   127
## 41 2015      5     1
## 42 2016      5     0
## 43 2017      5     0
## 44 2018      5     1
## 45 2019      5     0
## 46 2020      5     0
## 47 2021      5     0
## 48 2022      5     0
## 49 2023      5     2
## 50 2024      5     1
## 51 2015     NA    15
## 52 2016     NA    13
## 53 2017     NA    10
## 54 2018     NA    14
## 55 2019     NA    14
## 56 2020     NA     8
## 57 2021     NA     7
## 58 2022     NA     6
## 59 2023     NA     9
## 60 2024     NA     8
```

Obter variável raça/cor como palavras descritivas ao invés de números.

```
dados_nb_appended_melted <- dados_nb_appended_melted %>%
  dplyr::mutate(raca_cor_factor = ifelse(dados_nb_appended_melted$variable == "1", "br",
                                       ifelse(dados_nb_appended_melted$variable == "2",
                                             ifelse(dados_nb_appended_melted$variab
                                                    ifelse(dados_nb_appended_melted
                                                           ifelse(dados_nb_appended
                                                                ifelse(dados_nb_ap
```

Remove NA in year

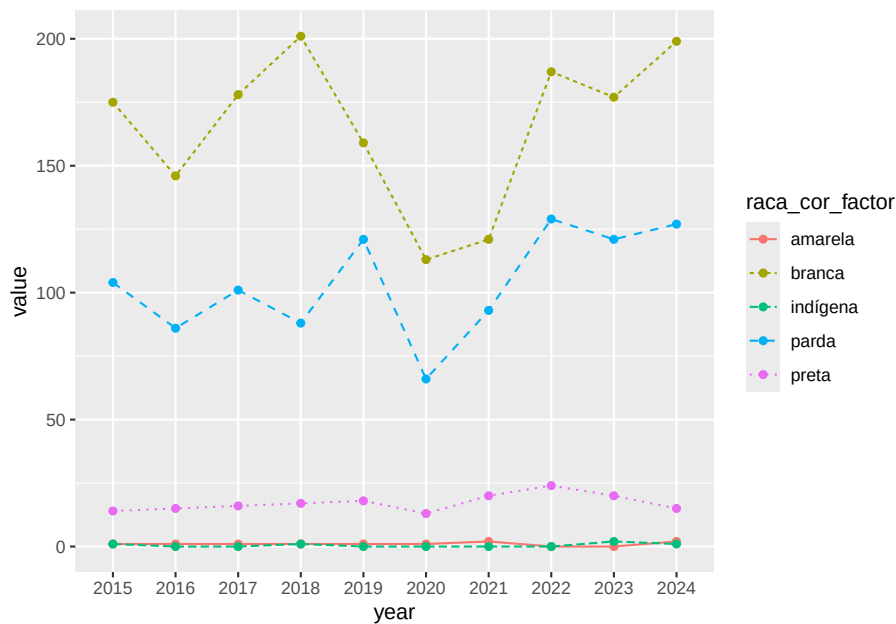
```
dados_nb_appended_melted <- dados_nb_appended_melted[ which(
  dados_nb_appended_melted$year != "NA"), ]
```

Remove 0 in raça/cor

```
dados_nb_appended_melted <- dados_nb_appended_melted[ which(
  dados_nb_appended_melted$raca_cor_factor != "0"), ]
```

Visualizar plot

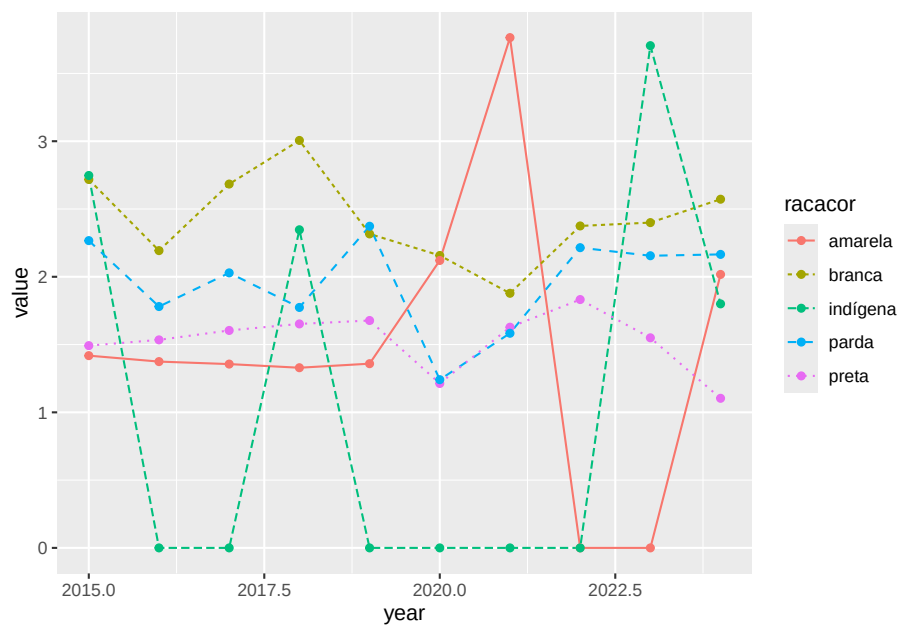
```
ggplot(dados_nb_appended_melted, aes(x=year, y=value, group=raca_cor_factor, color=raca_cor_factor)) +
  geom_line(aes(linetype=raca_cor_factor)) +
  geom_point()
```



Visualizar Plot Disparity

```
library('readxl')
Disparity_coefficient_Brazil_All <- read_excel("./data/Disparity_coefficient_Brazil_All.xlsx")

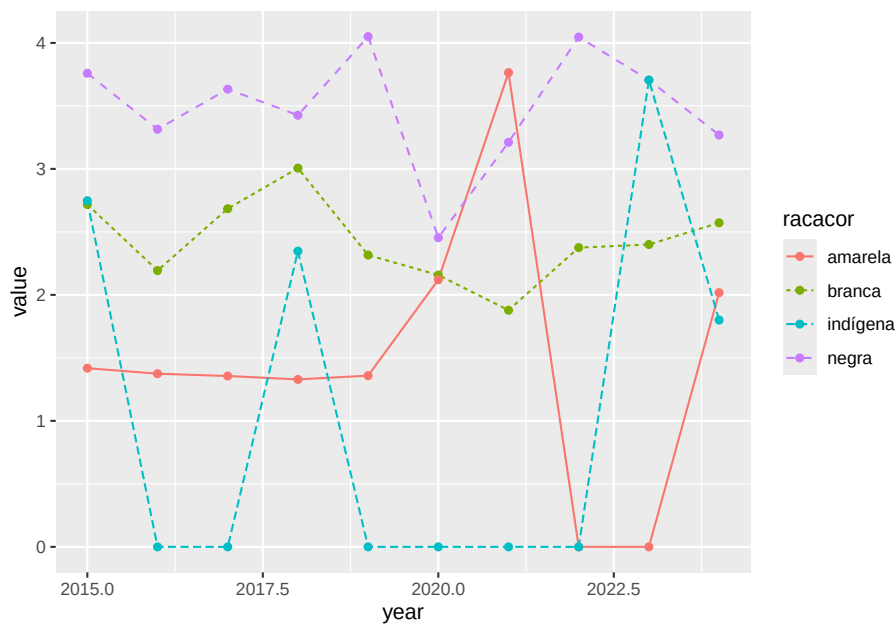
ggplot(Disparity_coefficient_Brazil_All, aes(x=year, y=value, group=racacor, color=racacor)) +
  geom_line(aes(linetype=racacor)) +
  geom_point()
```



Visualizar Plot Disparity

```
library('readxl')
Disparity_coefficient_Sum_PardosBlacks_Brazil_All <- read_excel("./data/Disparity_coefficient_Sum_PardosBlacks_Brazil_All.xlsx")

ggplot(Disparity_coefficient_Sum_PardosBlacks_Brazil_All, aes(x=year, y=value, group=racacor, color=racacor)) +
  geom_line(aes(linetype=racacor)) +
  geom_point()
```



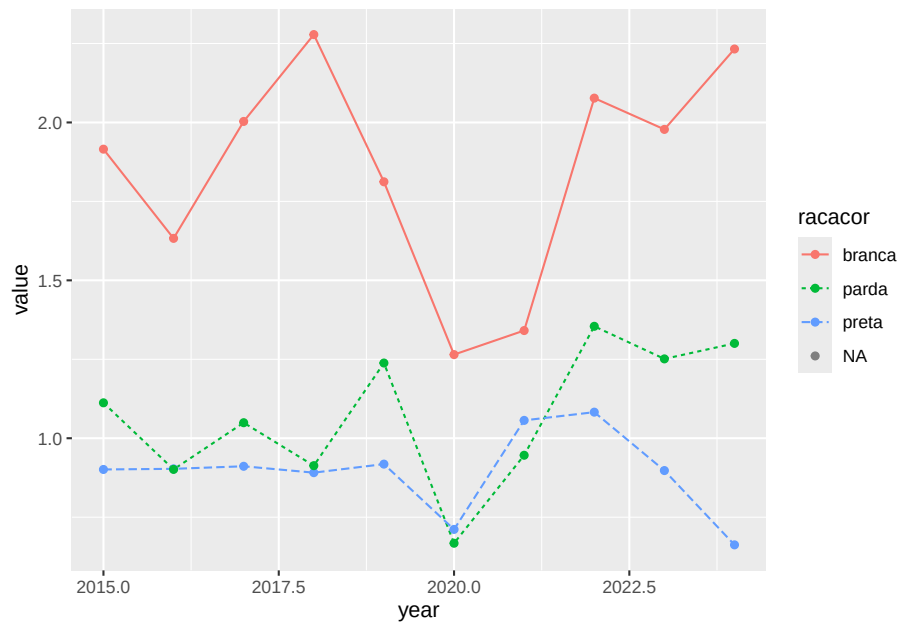
Visualizar Plot Disparity IBGE

```
library('readxl')
Disparity_ibge_plot <- read_excel("./data/Disparity_ibge_plot.xlsx")
```

```
ggplot(Disparity_ibge_plot, aes(x=year, y=value, group=racacor, color=racacor)) +
  geom_line(aes(linetype=racacor)) +
  geom_point()
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

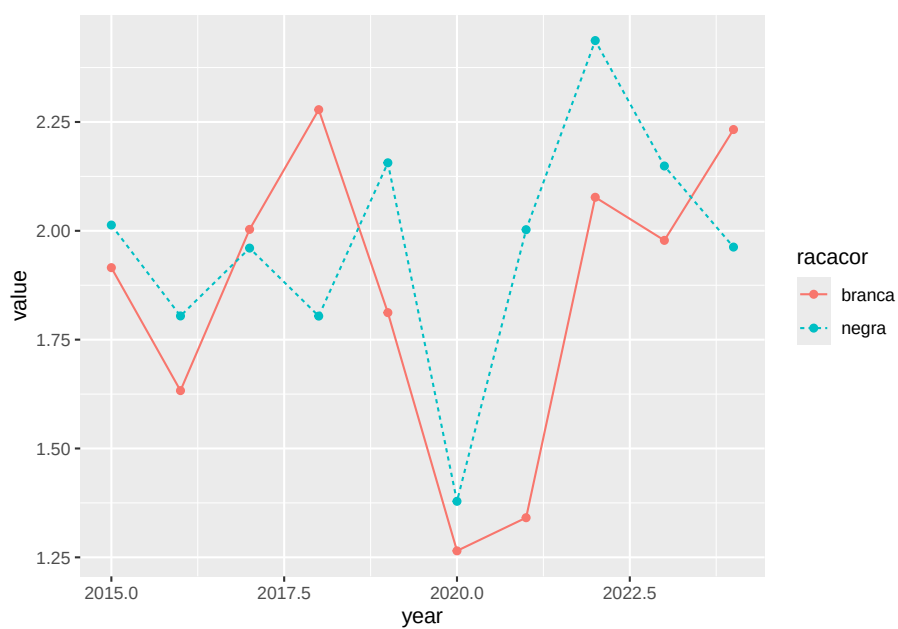
```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Visualizar Plot Disparity IBGE

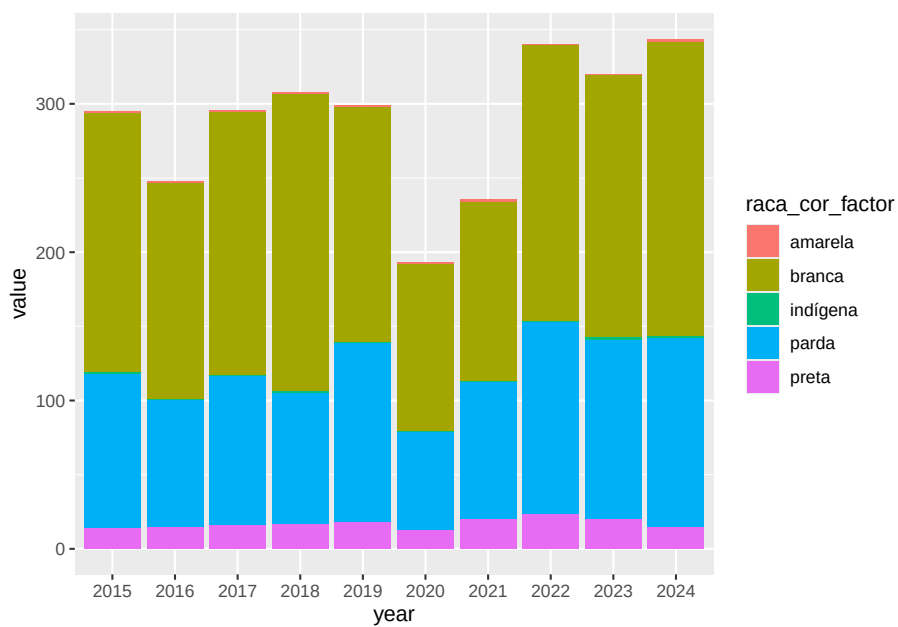
```
library('readxl')
Disparity_ibge_negros_plot <- read_excel("./data/Disparity_ibge_negros_plot.xlsx")
```

```
ggplot(Disparity_ibge_negros_plot, aes(x=year, y=value, group=racacor, color=racacor))
  geom_line(aes(linetype=racacor))+
  geom_point()
```



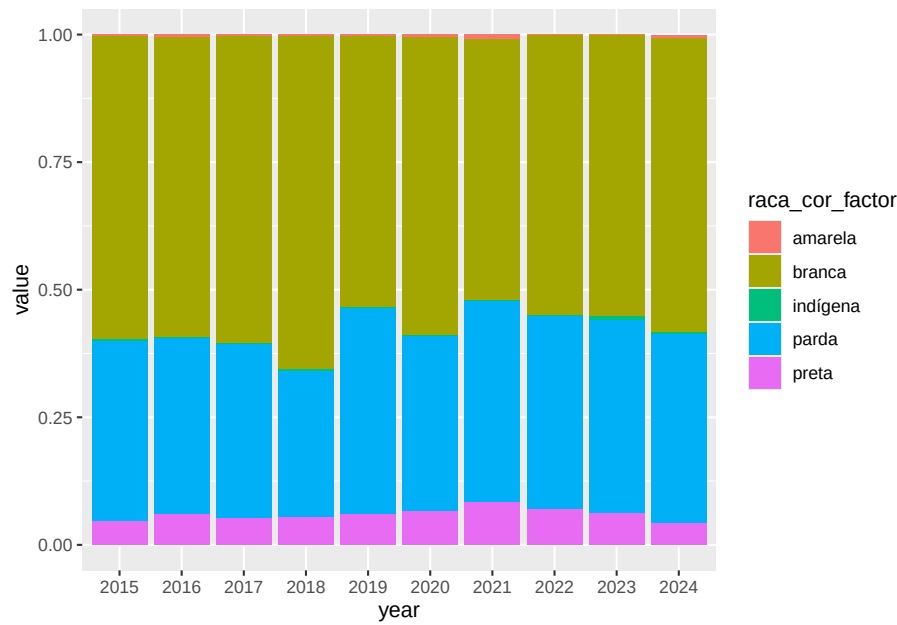
Stacked Graph

```
ggplot(dados_nb_appended_melted, aes(fill=raca_cor_factor, y=value, x=year)) +  
  geom_bar(position="stack", stat="identity")
```



Stacked + Graph

```
ggplot(dados_nb_appended_melted, aes(fill=raca_cor_factor, y=value, x=year)) +  
  geom_bar(position="fill", stat="identity")
```



Chapter 6

Final Words

We have finished a nice book.

Bibliography